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KENYATTA UNIVERSITY INSTITUTE OF OPEN LEARNING

EPS 200: DEVELOPMENTAL PSYCHOLOGY

**DEPARTMENT OF EDUCATIONAL
PSYCHOLOGY**

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PREFACE:

Human growth and development is the study of human beings from conception to death. It seeks to integrate the basic concepts and principles of physical, motor, cognitive, language, social, emotional, moral and personality development, each major stage of life – pre-natal infancy, childhood, adolescence, adulthood and old age. This course provides an essential background for students in psychology, nursing, education, social welfare, home economics, workers in community service and parents. The course is also intended to help you understand your own behavior, also provides insight into the behavior of others which should help you to achieve better relationships with those around you. The course is meant to help you understand the various theories, facts and concepts related to human growth and development.

GOALS OF THE COURSE

By the end of the course the student will be able: -

- To envisage a human being from pre-natal stage to old age.
- Understand the underlying principles of development.
- Conceptualize prenatal development and understand the importance of a healthy pregnancy.
- Define and explain the different behaviors of a neonate and an infant.
- Understand that a child in early, middle and late childhood undergoes an expansion and development of those behaviors of infancy.
- Appreciate the unique development of the adolescent.
- Understand the problems and challenges of the adolescent period.
- Appreciate the importance of studying adulthood development.
- To discover some consistent processes and major influences throughout the life span.
- To understand the influences of childhood on later adulthood.

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LESSON ONE: INTRODUCTION TO HUMAN GROWTH AND DEVELOPMENT

1.0 INTRODUCTION

In this introductory lesson, we are going to introduce you to the concepts of human growth and development. In the first part we will look at definitions of the terms growth and development, followed by the relationship between growth and development. In the second part we will look at the principles of growth and development and finally the importance of studying human growth and development.

OBJECTIVES



By the end of this lesson, you should be able to:

1. Define the terms growth and development.
2. Discuss the principles of growth and development.
3. Explain the importance of studying human growth and development.

1.0. DEFINITION OF TERMS

Growth

Growth refers to the physical and physiological changes that occur throughout life. These changes are quantitative such as physical growth (change in weight and height) and occur during the early phases of life up to adolescent years. Quantitative changes refer to measurable and obvious features of human development.

Development

Development refers to change in function and such changes are non-organic and usually qualitative, for example language usage, thought processing, reading skills, and peer relationships. Qualitative changes refer to the alterations in human functioning through the life span.

1.2. RELATIONSHIP BETWEEN GROWTH AND DEVELOPMENT

- 1) Development is achieved through growth. One can grow but not develop. For example a child can increase in size but may not be able to walk.
- 2) Development is a continuous life long process while growth is limited.
- 3) Growth refers to organic changes while development refers to non-organic – functional changes.
- 4) Growth is quantitative while development is qualitative.

1.3. PRINCIPLES OF GROWTH AND DEVELOPMENT:

Principles are general patterns that hold true in describing the way human being develops.

The main principles are given below: -

1. Development follows a definite orderly sequential and predictable pattern rather than random. For example a child sits, then stands and walks. The order cannot be reversed.
2. Development follows two directional trends: -
 - i. Cephalocaudal trend – development proceeds from the head to the heel. For example, the child can hold the head upright and use the hands before walking or using the legs.
 - ii. Proximodistal trend – development proceeds from the centerline of the body to the other parts. For example, the child can wave the hands before using the fingers to pick something.
3. Although development is a continuous process, it is not always smooth and gradual. There are growth spurts, that is periods of accelerated growth. For example, the first two years and the adolescent period are periods of accelerated growth.
4. There are critical or sensitive periods in the development of body organs and psychological functions. These are periods when the potential for growth as well as

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harm from the environment is at its maximum. Any interference may result in permanent deficiencies or malfunction, for example fourteenth to ninetieth day. This is the period when most of the organs of the body are developing.

5. All developmental changes are a product of two basic processes – maturation (heredity) and experience (learning).
6. Development is shaped by its historical/cultural context. For example those born in the age of computers and Internet are likely to have different experiences.
7. Different aspects of development are interrelated. For example physical development will influence social and mental development.
8. Human development is a holistic process and not piecemeal.
9. Development is an individualized process. That is, each child has a characteristic rate of development regardless of the stage of development. These differences may be due to genetic factors, nutrition, illness, lack of exercise, etc.
10. Development proceeds from general to specific and simple to complex. For example babies move their arms (gross movement) before grasping with thumb and index finger (specific).
11. Development is a cumulative process, i.e. changes do not emerge all of a sudden. Past events often have implications for the future.
12. There is much plasticity in human development. Plasticity refers to the capacity for change in response to positive or negative life experiences. The course of development can change abruptly if important aspects of one's life change.

1.4. IMPORTANCE OF STUDYING HUMAN GROWTH AND DEVELOPMENT

- 1) The teacher must know the basic principles of growth and development and the characteristics that emerge at different age levels in order to provide effective guidance for harmonious development of children.
- 2) Children come to school with individual differences due to their different homes and environment. The teacher needs to know the potentialities and capabilities of each and every child so that he may exploit them to the maximum for the benefit of society.
- 3) To ensure continuity from the past to the present, the teacher needs to understand past aspects of development. The teacher will understand the present in terms of its past. What students are is as a result of what they have been through.
- 4) The teacher is an agent of socialization and there is need to know those whom he will socialize in order to facilitate their development as good citizens.
- 5) The teacher needs to know and understand the basis of development and what is abnormal.
- 6) To improve the quality of learning and teaching because the teacher knows the normal characteristics of children at different ages.
- 7) The teacher will be able to understand himself/herself better.

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1.5. Summary



- ✓ Growth and development is both quantitative and qualitative.
- ✓ There are general patterns or principles of development.
- ✓ It is important to study human growth and development so as to understand others and ourselves around us.

Review questions



- 1) Compare and contrast growth and development giving examples.
- 2) Describe the basic principles of growth and development.
- 3) Explain why it is important to study human growth and development.

LESSON TWO: RESEARCH METHODS AND DESIGNS IN HUMAN DEVELOPMENT**2.0. INTRODUCTION**

In the previous lesson, we introduced you to the concepts of growth and development, the underlying principles of growth and development and why it is important to study the course. In this lesson, we are going to look at the research methods used to study human beings at various stages of development. We will also look at research designs and ethical standards for developmental psychology.

OBJECTIVES

By the end of this lesson the student should be able to:

1. Identify and explain the various methods and research designs used to study human growth and development.
2. Discuss the various ethical considerations in the study of human development.

2.1. RESEARCH METHODS AND DESIGNS

The following are the various methods of research in human growth and development: -

2.1.1 Observation

This is a data collection strategy in which the ongoing behavior of individuals is recorded with as little interaction between observer and subject as possible. The goal of this research is to determine exactly what individuals are doing in a specific setting. Observational techniques have the following advantages and disadvantages.

Advantages

- Observes subjects in their natural settings.
- Development can be observed in various stages.
- The researcher does not impose themselves on the subjects.
- Is important for determining how children and adults function in their everyday life.

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Disadvantages

- It does not answer the question of cause and effect (does not tell us how and why events are happening).
- It does not allow for experimental control to determine how research variables are related to one another.
- It does not tell us about the internal psychological events and experiences of subjects, that is their thoughts, believes, fantasies, etc.
- These procedures are inefficient when studying behavior occur infrequently or irregularly in the subjects' everyday life.
- One has to wait for along time before the behavior of interest is emitted.
- Problem of observer bias – one is likely to record the positive aspects and omit the negative ones or vice versa.
- Sometimes one may not be able to apply all the information to other subjects in general. That is the information may not be generalized to others.

2.1.2 Experimental techniques

This is a data collection strategy in which one set of variables (independent variable) are manipulated and their influence observed on a second set of variables (dependent variable). It is most appropriate when one wants to establish cause-effect relationships and can also be used to study behavior that occurs irregularly or infrequently in every day life. The procedure can be repeated or replicated by other investigators. It also provides a test for the reliability of findings. However, it has the following limitations: -

- Some behaviors are not open to experimental manipulation.

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- Some of the findings have limited applications – it tells what subjects can do in restricted laboratory conditions but not in real life.
- Artificiality of the experimental research for example sometimes subjects may try to look good and report better results than the case may be. Alternatively they may behave unnaturally or intentionally alter their behavior to sabotage the experiment.

2.1.3 Clinical methods

Involves a researcher and one individual or subject at a time. It combines observations and careful questioning. The clinical method is flexible. Questioning or the tasks given can be individualized for each person. The major disadvantage is that it may produce conclusions that are dependent on the interviewer ability to ask the right questions. In addition, results obtained may need to be verified by others.

To accomplish the goals of developmental research, scientists must discover how and why people change or remain the same over time. To accomplish developmental psychologists use two basic research designs.

2.1.4 Cross-sectional research design

It compares groups of individuals of different ages on the same measure at the same time. For example, studying moral reasoning of thirty and sixty year olds over a short period of time.

Advantages

- It is quick and relatively inexpensive.
- It gives a good overview of the phenomena under investigation.
- Studies a large number of subjects.
- Can retest at later date.

Disadvantages

- It does not tell us the historical antecedents of the behavior.
- It does not tell us about behavioral stability.

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- People from different age groups not only differ in chronological age but also in the time period in which they were born and raised (generational effects)

2.1.5 Longitudinal research design:

Is a research design in which a group of individuals is repeatedly tested over a period of time. This is meant to overcome the various limitations of cross-sectional designs. For example, testing same adults every ten years to see changes in their moral reasoning.

Advantages

- Provides a good picture of development within individuals.
- Can provide answers about developmental stability of behaviors.
- Can determine influences of earlier conditions or experiences.

Limitations

- Is time and money consuming – very expensive.
- Subjects may drop out due to death, illness, migration, wars, etc.
- The subjects who complete the study are likely to be more co-operative, motivated, more persistent and more competent – hence may be biased, making it difficult to draw valid conclusions.
- When subjects are tested repeatedly over long periods of time or they receive the same or similar tests more than once, they are likely to do better not because of developmental effects but cause of effects of repeated practice.
- Some changes in individuals may be due to the time of measurement rather than development.

2.1.6 Sequential research design (longitudinal/cross-sectional)

It combines cross sectional and longitudinal approaches. It starts with cross-sectional, then months or years later, you study the same individuals i.e. longitudinal aspects.

2.2. ETHICAL CONSIDERATIONS IN DEVELOPMENTAL RESEARCH

Research can expose subjects to possible physical or psychological harm. This brings us to the issue of research ethics. This section introduces you to some of the ethical considerations when conducting research. These include: -

1. Protection from physical or psychological harm, for example punishing children to study its effects.
2. Informed consent from the subjects themselves or guardians or parents. Involves giving correct information about the research, the purpose and operation of the research, etc. subject should be given the right to refuse consent.
3. Subjects should not be forced to participate in research.
4. Confidentiality – subject identity written or verbal reports and other informal discussions should be kept in strict confidence.
5. The researcher should report the general findings of the study to the subjects, parents and others.
6. Do not take advantage of the young and the old subjects. They have the right to be treated well.
7. Deception and debriefing – under some circumstances you can deceive the subjects, but you have to debrief them later.

2.3 Summary

- ✓ From this lesson we have seen several techniques developmental psychologists use to collect data (observation, experimental, clinical, etc). Each of these methods has their own strengths and weaknesses.

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- ✓ Various research designs are available to psychologists.
- ✓ Researchers need to be aware of ethical considerations.

Review questions and activities

- a) For each of the research methods discussed, formulate two problems that can be investigated with each method. For each of the problems, identify some of the ethical issues you would consider.
- b) Give two advantages and disadvantages, for the methods discussed.
- c) Compare and contrast longitudinal and cross-sectional designs.

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LESSON THREE: THEORIES AND DETERMINANTS OF HUMAN DEVELOPMENT

3.0. INTRODUCTION

In this lesson we are going to look at theories and determinants of development. The word theory means someone's guess about why something happens the way it does. According to Brodizinsky (1986) a theory is a set of coherent statements, laws and principles that describe, define and predict specific aspects of human development. The major purpose of examining theories of human development is to demonstrate how such theories are used to formulate answers to basic questions of human development.

We will also look at the main determinants of development. Both heredity and the type of environment in which one is raised influence human development.

OBJECTIVES



By the end of this lesson the learner should be able to:

1. Define what is meant by a theory.
2. Describe characteristics and functions of theories.
3. Describe the basic ideas and stages of the following theories:
 - Freud's psychosexual theory.
 - Erickson's psychosocial theory.
 - Behavioral theories.
 - Social-learning theories.
 - Piaget's theory.
 - Gessel's theory
 - Information processing approach.
4. Compare and contrast the above theories of human development and show how each account for development.
5. Understand the major determinants of human development.

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3.1. ROLE AND FUNCTIONS OF THEORIES:

Brodizinsky (1986) identified four roles of Theories: -

- Describes the conditions under which the phenomena being studied occur.
- Sets limits or boundaries determining the particular phenomena for which the theorist is responsible.
- Suggests possible relationships between theoretical constructs.
- Brings together an existing data into an integrated, logically consistent body of knowledge.

THEORIES OF HUMAN DEVELOPMENT

3.2.1. Gesell's Theory Of Maturation

According to this theory, development is guided by the unfolding of the individual's unique genetic plan. Children undergo growth and development at their own individual rates, i.e. human beings are programmed to develop according to a particular pattern. Environment hereby allows genetically programmed behavior to proceed in an orderly fashion. Therefore, there is a need for parents and caregivers to understand these individual rates of maturation. Don't push too much (concept of readiness). The major weaknesses are that it underestimates the role of environment and give little room for intervention.

3.2.2. Psychoanalytic Theory of Sigmund Freud

According to this theory: -

- i. Behavior is motivated by unconscious thought memories and feelings.
- ii. Life is the unfolding of sex instinct.
- iii. Child early experience affect later personality and mental health.
- iv. Children develop through a sequence of stages.
- v. People protect themselves from anxiety and other negative emotions through defense mechanism.

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Personality structure – According to Freud, personality has three constructs, id, ego, and super ego.

Id- the source of all wishes and desires.

- It's unconscious.
- Is seeks immediate gratification.
- Is like an infant. It operates on the pleasure principle.

Ego- principle – satisfies needs through socially appropriate manner.

- Delays gratification of id. Operates on the reality principle.

Super ego

- Inhibits Id's antisocial desires causing an individual to experience guilt.

It is the moral arm of personality. It represents our conscience. Some people cannot control their Id and thus cannot delay gratification or restrain their desires. Those people with weak super egos may not act in an ethical manner. When ego cannot balance between id and super ego it leads to anxiety, which results in the use of defense mechanisms. Defense mechanisms are automatic or unconscious processes that serve to relieve or reduce feelings of anxiety or emotional conflicts. These include repression, rationalization, reaction formation, regression, sublimation, denial, compensation, projection, and displacement.

Psychosexual stages

Freud says that the child comes into this world equipped with the libido or sexual energy. This sexual energy is satisfied in different parts of the body as the child grows. This forms the stages of personality development, which are as follows: -

1. Oral stage (0 – 1 years)

Sexuality is centered on oral cavity. Pleasure is obtained through sucking or biting. Frustration or over stimulation leads to fixation, which is reflected in ones personality. A person may expect to be mothered, sarcastic or smoking.

2. Anal stage (2 –3 years)

Sexuality is centered on the anal cavity. Pleasure is got through elimination or retention of waste products. The emphasis of parents is on toilet training. If toilet training is done smoothly, the child develops self-confidence, becomes creative and productive.

If parents are too strict and there is a lot of friction, the child learns to retain feces. Later in life the child may develop a retentive character, be stingy, aggressive, excessively stubborn and excessive cleanliness.

3. Phallic stage (4 – 5 years)

Sexual energy is attached to genitals. Oedipus and Electra complexes arise. In Oedipus complex, the boy starts to have sexual feelings for their mother but due to fear of being castrated by the father they identify with the father. On the other hand in Electra complex, the girl has feelings for their father but due to fear of their mother identify with the mother.

The process of identification is important because it makes the boy adopt the appropriate sex roles and behavior. If proper identification does not occur, the result will be poor development of sex roles and lack of conscience.

4. Latency (6 –11 years)

Child sexuality is dormant. Children spend time playing with same sex peers.

5. Genital stage (12 –19 years)

At this stage adult heterosexual behavior develops. Libido is re-awakened and a more mature sexual attachment occurs. Sexual objects are people of the opposite sex, first with adults e.g. teachers, older idols and later with peers. If confused identifications, one cannot cope with aroused sexual feelings at this stage. Also if oral period was not satisfactory, one will not have the foundations for basic love relationships.

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NB:

This theory shows the importance of early childhood experiences. There is need for subsequent experiences to ease the negative effect of poor early childhood experiences.

3.2.3. Psychosocial Theory of Eriksson

Eriksson believes that human beings face eight major crisis or conflicts in their lives. These form the stages of personality development. Each stage presents one with a crisis, if the crisis is well handled a positive outcome is met, where else if the crisis is not well handled a negative outcome is generated. Resolution of one stage brings the foundation for negotiating challenges of the next stage.

The table below presents the eight stages: -

Table 3.1: Erikson's stages of personality development

Age Range	Stage	Description
Birth to 1 year	Trust versus mistrust	Infants must learn to trust their caregivers to meet their needs. Responsive parenting is critical.
1 to 3 years	Autonomy versus shame and doubt	Children must learn to be autonomous – to assert their wills and do things for themselves or they will doubt their abilities.
3 to 6 years	Initiative versus guilt	Preschoolers develop initiative by devising and carrying out bold plans, but they must learn not to impinge on the rights of others
6 to 12 years	Industry versus inferiority	Children must master important social and academic skills and keep up with their peers or they will feel inferior.
12 to 20 years	Identity versus role confusion	Adolescents ask who they are and must establish social and vocational identities or else remain confused about the roles they should play as adults.
20 to 40 years	Intimacy versus isolation	Young adults seek to form a shared with another person but may intimacy and experience loneliness and isolation.
40 to 65 years	Generativity versus stagnation	Middle-aged adults must feel that they are producing something that will outlive them, either as parents, workers or they will become stagnant or self centered.
65 and older	Integrity versus Despair	Old adults must come to view their lives as meaningful in order to face death without worries and regrets.

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3.2.4. Piaget's Theory of Cognitive Development

According to Piaget, children think and deal with problems differently from adults.

Maturation, experience, social transmission and equilibration – that is a balance between what is known and what is being experienced, influence development.

The table below gives a summary of Piaget's stages.

Table 3.2: Piaget's stages of cognitive development.

Stage	Description
Sensory motor (birth to 2 years).	Learning is through the senses. Infants use sensory and motor activities to understand the world.
Pre-operational (2 to 7 years)	They can use symbols (words) to understand the world. Their thinking is egocentric. Immediate perception.
Concrete operational (7 to 11 years)	At this stage they can conserve, mentally classify and act on concrete objects.
Formal operational (12 years and older)	They can think about abstract ideas, hypothetical situations and systematically test hypothesis.

3.2.5. Behavioral Theory (Skinner, Watson)

According to this theory, human behavior is the result of environmental stimulation i.e. accumulated effects of learning. Much of what an individual becomes is the result of what he or she has experienced or learned. This can be through classical conditioning or operant conditioning. In classical conditioning, learning takes place through association, while in operant conditioning, learning depends on rewards and punishments.

3.2.6. Social Learning Theory (A. Bandura.)

Much of what we learn and how we develop occurs through observation and imitation of those around us.

3.2.7. Information Processing Theory

The main focus for the information processing theory is on the quantitative changes that occur in people's knowledge, skills and the efficiency with which we attend to, make sense of and remember information in the course of development. This theory is concerned with the changes in the way people transfer information, from one period of life to another, i.e. people become increasingly aware that they need to do something special in order to remember things. These transitions are not presented in stage like fashion like those in Piaget's theory. Instead it shows the flow of information beginning with input (stimulus) leading to some sort of output (response). This theory attempts to specify mechanisms by which people make transitions in their problem solving capacity.

3.2.8. Eclectic Approach to Understanding Behavior

One is likely to ask which is the best or correct theory. There is no one correct or right theory. Each theory defines its own area of investigation and its research methods. The theories differ from each other according to what are the most important aspects of behavior you want to study. Therefore each theory is true to its own area of inquiry.

We have to use the eclectic approach to understand behavior by picking and choosing elements from the different theories.

3.3. DETERMINANTS OF DEVELOPMENT

The question of what determines development is one of the oldest controversies in human history. The debate is on the relative impact of hereditary (nature) and environmental influences (nurture) in shaping personality traits and characteristics. Nature refers to the genetic influences in development, while nurture refers to the environmental influences on development, from prenatal to death. Information carried by genes in chromosomes influences the sequence of growth and maturation, timing of puberty, course of aging, eye color, skin color, susceptibility to some diseases, intelligence, temperament, personality traits, (aggressiveness, smiling, responses, fear, shyness). It also influences the sequence

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of psychological disorders (depression and schizophrenia), alcoholism, body size and shape, height, athletic potential.

Nurture includes all the individual's experiences in the outside world i.e. the family, school, community, and the culture at large. Both interact to produce a human being without genes. A person without environment is no person. The interaction between nature and nurture is the crucial influence on any particular aspect of development.

3.4 Summary



- ✓ A theory is a set of understandings about development, maturation, and learning.
- ✓ There is no one true, correct, right theory. Each provides a window of understanding development.
- ✓ Psychoanalytic theory focuses on early childhood experiences and unconscious forces.
- ✓ Piaget's cognitive theory addresses cognition and changes in the structure in the thinking, from infancy to adolescence.
- ✓ Behaviorist theory looks at the role of environmental factors in development.
- ✓ Social learning theory focuses on the role of observation and initiation.
- ✓ The information processing theory describes the way in which we attend to, make sense of and remember information during the course of development.
- ✓ Maturational biological theory emphasizes the importance of the individual's rate of maturation.
- ✓ Human development is determined by the interaction between heredity and aspects of the physical and social environment.

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Review Questions



- 1) Discuss the main characteristics of each of the theories showing how each theory account for development.
- 2) Discuss the major environmental influences on human development in your local community.
- 3) Discuss how genes influence one's physical characteristics of behavior.

LESSON FOUR: PRE-NATAL DEVELOPMENT**4.0. INTRODUCTION**

In this lesson we will look at human development during the prenatal stage of development. Specifically we will look at the stages of prenatal development and the social and environmental factors that affect development during this period. We will also look at the birth process and how it may affect the child.

OBJECTIVES

By the end of the lesson the students should be able to:

- Describe the periods of prenatal development.
- Explain the major features of each period.
- Identify and explain the major environmental influences (teratogens) of prenatal period.
- Discuss the problems that occur during birth and their solutions.
- Identify critical periods in development.
- Identify and describe major biological risk factors during the prenatal period.

4.1. STAGES OF PRENATAL DEVELOPMENT

Human life begins at conception, which occurs when a sperm cell unites with an egg from a female and forms a single cell called a zygote.

After conception the growth and development of the human being proceeds as a result of cell division. There are two processes of cell division: -

- Mitosis – is the process of cell division in which a single body (somatic) cell divides into two exactly equal parts. Each of the parts has exactly the same 23 pairs of chromosomes as in the original cell.
- Meiosis – is the process of cell division that occurs during the production of sex or germ cells (ova and sperm). During this process each chromosome splits and

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separates so that the resulting ovum or sperm contains only 23 single chromosomes.

The period between conception and birth is referred to as the prenatal period. Human growth and development occurs most rapidly during this period. Prenatal development is divided into three stages: -

1. Germinal period/pre-embryonic period (conception -14 days)

This period is characterized by rapid cell division and ends when the zygote implants itself in the walls of the uterus.

2. Embryonic stage (3-8 weeks)

During this stage, cell differentiation occurs in which the cells of the blastocyst differentiate themselves into three layers. This is followed by organogenesis in which different organs of the body begin to form. Other developments during this period are the formation of the amniotic sac – which is filled by amniotic fluid. The placenta is also formed through which the embryo gets its nutrients during pregnancy. By the 8th week, the embryo has rudimentary eyes, ears, nose, mouth, liver, heart, and webbed fingers. This period is very critical for development. This is because:

- The embryo is most susceptible to influences of teratogens – viruses, chemicals, drugs and radiation. If it is exposed to any of these, it is most likely that the embryo will have abnormalities of the organs of the body, which are in the process of being formed.
- Most miscarriages occur at this period. Embryo detaches from the wall and is expelled.

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3. Period of the fetus (9-40 weeks)

During this period, the major systems begin to function. There is also refining and improving of what is already formed. There is a major growth and development of the brain and the nervous system during the last three months of prenatal development. The 28th week marks the point of viability, which is the point at which survival outside the uterus may be possible. The baby is born approximately 266 days after conception or 280 days from the onset of the last period.

Table 4.1: Summary of prenatal milestones

Event	Period	Time span	Name of organism
Implantation	Germinal	2 weeks	Zygote
Organs form	Embryonic	3-2 months	Embryo
First heart beat	Embryonic	40 days (6 weeks.)	Embryo
Kicks first felt by mother (quickening)	Fetal	16 th week	Fetus
Can tell sex of child	Fetal	12 th week	Fetus
Age of viability	Fetal	28 th week	Fetus

4.2. FACTORS AFFECTING PRENATAL DEVELOPMENT

Each year some babies are born with birth defects that range from gross anomalies that may lead to sudden death to minimal physical or mental defects. About 7% of children are born with a congenital defect. These defects may be caused by the following factors: -

- Abnormal genes and chromosomes.
- Harmful conditions in the womb.
- Abnormalities during the birth process.

4.2.1. Abnormal genes and chromosomes

- (a) Genetic abnormalities: - genetic abnormalities result from mutations – that is, a change in the chemical structure or arrangement of one or more genes that has the

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effect of producing a new phenotype. Some conditions that may result from this are: -

- (i.) Cystic fibrosis – occurs in about 1 in 1000 births. The child lacks an enzyme that prevents mucus from obstructing the lungs and the digestive system. Many who inherit this problem die in childhood or adolescence.
 - (ii). Muscular dystrophy (MD) – it attacks the muscles. As the disease progresses, the individual starts showing slurred speech, becomes unable to walk, and may gradually lose most of the motor capabilities.
 - (iii). Phenylketonuria (PKU) – the child lacks an enzyme to digest foods that contain the amino acid phenylalanine (including milk). If not detected and the child placed on a diet of milk substitute, phenylpyruvic acid will accumulate in the body and attacks the developing nervous system. Long term effects of PKU are hyperactivity and severe mental retardation.
 - (iv). Tay-sachs disease is a degenerative disease of the nervous system that kills the victim by the 3rd birthday.
 - (v). Hemophilia (bleeder's disease) – the child lacks a substance that causes the blood to clot and can bleed to death if cut. Is rare in females.
 - (vi). Diabetes – an inherited condition in which the individual is unable to metabolize sugar properly because the body does not produce insulin. Can be controlled by taking insulin and restricting ones diet.
- (b) Chromosomal abnormalities: - usually occur during the process of cell division after conception (meiosis and mitosis). May be caused by non-separation of chromosomes, deletion of chromosomes which may lead to loss of genes, and translocation of chromosomes i.e. deleting and attaching themselves to the wrong

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ones that is uneven segregation of chromosomes. Some conditions that they result include: -

- Turner's syndrome – XO – the person is female but is short, mentally retarded and sterile.
- Klinefelter's syndrome – XXY – a male with male external organs but with female body contours. Also sterile.
- Downes syndrome (mongolism) – the 21st chromosome does not separate so the zygote ends up having two no.21 chromosomes (trisomy 21). The child is mentally retarded and may have congenital eyes, ear, and heart defects; sloping forehead, protruding tongue, short stubby limbs, a slightly flattened nose, a distinctive fold to the eyelids that gives their eyes an oriental appearance.

N.B: The probability of having a child with Downes syndrome and klinefelter's syndrome increases dramatically if the mother is over 35 years. These defects can be detected in the first three months of pregnancy through a test called amniocentesis.

4.2.2. ENVIRONMENTAL INFLUENCES (TERATOGENS)

A teratogen is an external agent that may adversely affect the baby if the mother is exposed during the prenatal period. In this section, we will look at some of the teratogens.

1. Maternal characteristics: -

- (i). Maternal age - young mothers that are those below 18 years are more likely to die at birth, have a stillborn and neonatal death. This is because they are not mature enough, and do not receive prenatal care. Older mother over 35 years – may have small infants; premature, stillborn and Downs syndrome.

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- (ii). Mother's emotional state: - if the mother is, depressed, resentful, and it lasts throughout pregnancy, it can lead to miscarriage, prolonged or painful labor. The babies will tend to be hyperactive irritable and have irregular sleeping, feeding, and bowel patterns.
- (iii). Mothers diet: - retarded fetal growth, malformations, less developed brain, greater vulnerability to disease, low birth weight, cerebral palsy and epilepsy.
- (iv). Maternal diseases. Table 4.2 below shows some of the maternal diseases and the possible effects on the mother.

Table 4.2: Maternal diseases and their effects.

Disease	Effects
Rubella	Blindness, deafness, heart abnormalities and mental retardation
Syphilis	miscarriage; serious eye, ear, bone and brain damage
Gonorrhea	Blindness.
Genital herpes (herpes complex)	Neonatal deaths; blindness; brain damage and other serious neurological disorders. It has no cure. Victims advised to undergo a caesarian delivery.
Rhesus diseases	Incompatibility between Rh-negative mothers and Rh-positive fetuses. Does not affect first child. Can be controlled by administering Rhogam- a drug that prevents the mother from forming rhesus antibodies that could harm her next rhesus positive child.
Hypertension	miscarriages and infant death
Influenza	Powerful strains can induce spontaneous abortion and a number of abnormalities during the early stages.
Fetal death	Miscarriage and stillborn.
AIDS	Child born with AIDS.
Toxoplasmosis	a mild disease that produces symptoms similar to cold. It is caused by a parasite present in raw meat and cat faeces. It produces serious eye and brain damage and possible death of the unborn child.
Smallpox	Risk of miscarriage and stillbirth.

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(v). Drugs:

The table below shows some of the drugs and their possible effects.

Table 4.3: Drugs and their effects during the prenatal period.

Thalidomide. Malformation of limbs, eyes, ears and heart.
Sex hormones. For example pills which may result to heart defects and diethylstilbestrol (DES) – a synthetic hormone prescribed to prevent miscarriage. It can produce cervical cancer in adolescent female offspring and genital. Sex hormones can result to tract abnormalities and sterility in males.
Alcohol – fetal alcohol syndrome (FAS); microcephally, malformations of the heart, limbs, joints and face.
Tobacco – spontaneous abortion, stillbirth and infant mortality. Also low birth weight.
Antibiotics – for example streptomycin, terramycin and tetracycline can lead to hearing loss, prematurity, retarded skeletal growth, cataracts and staining of babies’ teeth.
Aspirin – large quantities may cause remittal bleeding and gastrointestinal discomfort.
Narcotics – growth deficiency, withdrawal syndrome, respiratory depression, death.
Barbiturates – large doses lead to anoxia; interfere with baby’s breathing.
Vitamins –excessive vitamin A can cause cleft palate, malformed heart and other serious birth defects.
Radiation – malformation and cancer.
Quinine – large doses cause deafness.
Chemicals – dyes, food additives, paints and pollutants – if mother is exposed, may affect the baby.

4.3. THE BIRTH PROCESS

Although a baby may be healthy during pregnancy, some things may happen between the beginning of labor and the birth of the baby, which may affect the baby. These are:-

1. Anoxia – i.e. lack of sufficient oxygen is due to: -

- Prolonged labor
- Failure to breathe when separated from mother’s supply of blood.
- Use of painkillers during birth.
- Too much anesthesia.

Consequences:

May include brain damage, death, and cerebral palsy – characterized by paralysis of legs or arms, tremor of the face or fingers, and inability to use vocal muscle.

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2. Use of instruments: forceps and vacuum extraction may be used when baby is too big or the mothers' is not ready - well developed to deliver baby.

Consequences:

Put strong pressure on the baby, blood vessels may break – hemorrhage in the brain of the baby. This results to brain damage, epilepsy, death, and cerebral palsy.

3. Unhygienic conditions: May infect the baby for example tetanus.

4.4. Implications

What can be done to ensure growth and development of healthy children?

1. Genetic counseling and screening – can help a couple predict whether they will have a child with a genetic problem. Can be done by checking family history and background. If there is a risk can go for amniocentesis, chorionic villus sampling (CVS), and ultrasound.
2. Treatment and control of hereditary disorders e.g. PKU – put on special diet, Hormonal treatment for turners and klinefelters syndrome.
3. Get children between 18 and 35 years
4. Good pre-natal care and immunizations.
5. Pregnant mothers to avoid exposure to contagious diseases e.g. gonorrhea, herpes, syphilis, toxoplasmosis.
6. Pregnant women to avoid taking any drugs unless they are absolutely essential and only when prescribed by a physician. Also avoid radiation treatments and x-ray examinations and no smoking during pregnancy.
7. A prospective mother who is rhesus negative should make sure her physician takes the necessary steps to protect the unborn baby and all subsequent children from Rh disease. Can be controlled by administering Rhogam – a drug that prevents the mother from forming rhesus antibodies that could harm her Rh-positive child. Does not affect first child.
8. A nourishing diet, rich in proteins and adequate in calories is essential during pregnancy.

9. Proper medical care during birth and hygienic conditions.

4.5. Summary



- ✓ The process of conception occurs when fertilization takes place.
- ✓ There are three stages in pre natal development germinal stage, embryonic and fetal stage.
- ✓ Genetic and chromosomal abnormalities may have negative impact on the developing fetus.
- ✓ Some prenatal environmental factors that may affect the development of the human being include the following; maternal nutrition, rhesus factor incompatibility, maternal age, maternal emotional status, drugs and maternal diseases.
- ✓ The birth process may be marked by birth hazards that may have far reaching consequences for example prolonged labor, anoxia and mode of delivery.

Review questions and activities

1. Discuss cultural practices and traditional beliefs in your community and their effects on the growth and development of a child during pregnancy.
 2. What can be done to ensure growth and development of healthy children during pregnancy and birth.
 3. How can we best meet the basic needs of babies before they are born?
 4. Home and hospital births, what are your recommendations and why.
- What factors affect prenatal development?

LESSON FIVE: THE NEONATE**5.0. INTRODUCTION**

In this lesson, we will study the neonate, i.e. a newborn baby up to about four weeks. We will look at their physical appearance, their behavioral capacities which include sensory and reflective behaviors. Finally we will look at the developmental needs that have to be met to ensure a healthy development of the baby.

OBJECTIVES

By the end of this unit, the learner should be able to

1. Describe the physical appearance of a neonate.
2. State some behavioral capacities of a neonate.
3. Describe the test used to ascertain the normality of a neonate.
4. State important needs necessary for the growth and development of a neonate.

5.1. Physical appearance and tests used to ascertain normality

The average weight of a newborn is 7 to 7.5 lbs or 3.3 kgs. Average length is 20 in or 51 cm. The head is normally about a quarter of the total size. Many newborns may look abnormal and ugly. They are wet, sticky and brown/black. The skin looks wrinkled; their heads are misshapen – oval like. They have flattened noses, the eyes look bruised and swollen. The skin is normally coated with a cheese – like protective substance called **vernix caseosa**. When it dries it leaves the baby with a chalky like appearance.

A soft fine hair that covers the fetus body during the 5th and 6th months of prenatal development called **lanugo** may persist on some parts of the newborn body for a few weeks after birth. The neonate has six soft spots called **fontanelles** on the head. The most conspicuous is the one on top of the head. It closes at about 1.5 years. Many neonates have enlarged breasts that secrete milk like substance. The girls may have a brief

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menstrual flow caused by hormones absorbed from the mother's blood. The hair is dark and soft while others are born bald.

To find out whether the neonate is normal and has no problems at birth, the Apgar score is taken at one minute and five minutes after birth. The attendant observes the pulse, breathing, muscle tone, general reflex responses and the color of the skin (for blacks the mucous membranes, palms or soles). A perfect Apgar score is 10 points; 7 or more is considered normal; below 7 shows that some bodily processes are not functioning fully and require watching and special attention. A score of 4 or less requires immediate emergence measures. Table 5.1 below illustrates how this is done.

Table 5.1: The Apgar scoring system of infants.

	0	1	2
Pulse	Absent	Less than 100	More than 100
Breathing	Absent	Slow/irregular	Strong cry
Muscle tone	Limp	Some flexion of extremities	Active motion
Reflex responses	No Response	Grimace	Vigorous cry
Color	Blue/pale	Pink on body proximity's/ blue	Completely pink

5.2 BEHAVIORAL CAPACITIES OF THE NEONATE

5.2.1 Reflexive Behaviors:

A baby is equipped with reflexive behaviors at birth. A reflex is a response that is automatic and is triggered involuntarily by some specific stimulus. A baby is born with some essential reflexive behaviors. Some of these reflexes help to ascertain that it is normal. These are: -

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1. The Rooting Reflex

When the baby is touched on the cheek, it turns its head and root around to put its mouth on the object that touches it. This reflex helps in finding the nipple. It disappears at 3 to 4 months.

2. Sucking Reflex

It begins in the uterus and it is essential for the baby's survival. The baby shows rhythmic sucking when anything for example a finger or nipple is inserted in the mouth. It disappears 3 to 4 months and the baby begins to suck selectively.

3. Swallowing Reflex.

The neonate is born with the swallowing reflex but it does not stop breathing so it swallows a lot of air, which has to be burped out.

NB: The above 3 reflexes are essential for the child's survival. They are called **feeding reflexes**.

4. The Moro Reflex

When startled, e.g. by a loud sound, the baby reacts by extending both arms to the side, with fingers outstretched as if to catch onto someone or something. The arms are then brought back to the midline as if they are hugging themselves. It disappears within 6 months.

5. Babinski Reflex

If the baby is stroked on the bottom of the foot, it will first spread out the toes, then curls them in. An older child curls them in immediately. If it occurs in an older child or an adult, it is a sign of abnormality in the neurological system. Should disappear by the end of first year.

6. Palmer grasp /Grasping Reflex.

If the baby is touched across the foot or palm, the fingers or toes close tightly over the object in a very firm grip such that the baby can hang on it.

NB: Reflexes 4, 5, and 6 are called **primitive reflexes**. The mid -brain, i.e. the part of the brain that develops first, controls them. They usually drop out as the brain develops. If they persist in life, it is an indication of some mal-development or damage to the brain.

7. Blinking Reflex

The eyes close if light is flashed or air is puffed across the eyes. This is a permanent reflexive behavior.

8. Knee Jerk

There is a rapid extension of the leg and knees if the kneecap is tapped. This is a permanent reflex.

9. Stepping Reflex

When newborns are held vertically, with their feet against a hard surface, they lift one leg away from the surface, and if tilted slightly from one side to the other, they appear to be walking.

10. Diving Reflex

The heart rate slows down when cold water is splashed on the face of the baby. This reflex becomes weaker with age.

11. Swimming Reflex

5.2.2 Sensory Capacities of the Neonate

Babies have all sensory abilities at birth.

1. Hearing – Babies hear since they are startled by a loud sound. They generally respond to higher and not lower sounds. They can locate sounds by 6 months.
2. Vision – Neonates eyes are sensitive to brightness. The pupils contract in bright light and dilate in darkness. By 2 weeks, the baby cries if they see a rapidly approaching object. They have some control over eye movements. They are not able to focus both eyes on the same point until the end of the 2nd month. They have a rudimentary ability to move both eyes in the same direction. They are able to focus only on close objects, usually 7 to 20 inches away – mother’s face. They probably do not see colors at all, only black, white and gray. They begin to respond to colors around the 3rd week. They prefer to look at complex patterns. They are most sensitive to the edges and contour of objects especially curves – called contour scanning. The baby is unable to see well at birth because:
 - a. Their eyes are smaller than those of adults so they receive images on fewer cells.
 - b. The protective myelin sheath covering the nerve fibers that transmit images to the brain is not yet formed. This sheath is thought to insulate the nerve fibers and insure that impulses traveling along one nerve path do not “leak” out to another.
3. Smell and Taste – These are fully operational. They react with some distress to unpleasant odors like ammonia and rotten eggs. They can differentiate between sweet, salty, sour and bitter tastes. They will grimace.
4. Touch – Is highly developed as they respond to the reflexes. When held they stop crying. They are initially not very sensitive to pain. If temperature is too low or high, they will respond by showing irritability and even crying.

5.3 OTHER SKILLS OF THE NEONATE

1. Learning – At 4 days, babies are able to learn through classical conditioning. An experiment by Lipsitt and Kay – sucking reflex (UCR & CR); nipple (UCS); and tone (CS).
2. Habituation – Reaching a stage of not responding to a stimulus and not hearing it. Results automatically from repeated exposure to the same stimuli.

Baby's Cycles

The neonate must establish clear regular cycles of sleeping and waking. Failure shows a sign of trouble. REM – rapid eye movements, i.e. a fluttering of the eyeballs under the closed lids, characterizes a large percentage of sleep time. A typical neonate cycle is sleep – wake – eat – alert – drowsy – sleep.

Limitations of the Neonate

1. Lacks vocal sounds save crying
2. Cannot raise head, roll over, or move thumb and fingers separately.
3. Cannot reach out at things at a distance
4. Cannot control elimination
5. Cannot differentiate one face from another
6. Cannot remember.

Sex Differences at Birth

1. Girls are smaller and lighter than boys but are a month to 6 weeks ahead in body development.
2. More boys have physical abnormalities than girls. They score low on Apgar scores at birth. 33% of boys die within the first year

Individual Differences at Birth

Babies differ in the following aspects due to their genetic make up and conditions in the womb:

1. Vigor of responding
2. General activity rate
3. Sleeping rhythms
4. Irritability
5. Rate of habituation
6. Cuddliness
7. Sensitivity to light or sound
8. In size and shape
9. In absolute and relative size of body organs
10. In hormonal balance
11. In blood chemistry

5.4 Summary



- ✓ After the birth of a baby, the Apgar neonatal scale is used to ascertain normality of the baby.
- ✓ The baby is born with a number of reflexes, which helps it to survive and help to assess its developmental level.
- ✓ Most of the babies' sensory capacities are functional at birth.
- ✓ The following physiological and psychological needs have to be met; food, water, air, proper temperature, love and affection, praise and recognition, new experiences and intellectual stimulant, activity and rest, security.

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Review Questions



1. Identify cultural practices and traditional beliefs in your community during birth and the neonatal period and discuss how they may influence the growth and development of children.
2. To what extent are the developmental needs met in your community.

LESSON SIX: PHYSICAL AND MOTOR DEVELOPMENT**6.0 INTRODUCTION**

In this lesson, we will discuss physical and motor development from infancy to the adolescent stage. The following aspects will be considered: - changes in height and weight; development of gross and fine motor skills and factor that influence physical motor development.

OBJECTIVES

By the end of this lesson the learner should be able to:

1. Discuss changes in weight and height as indicators of growth in childhood and adolescence.
2. Explain the process of brain, skeletal and muscular development
3. Discuss the development of motor skills at various ages
4. Discuss the role of learning and maturation on physical and motor development.

6.1 CHANGES IN HEIGHT AND WEIGHT

At birth the head appears too big when compared to the rest of the body. Body shape changes following the Cephalocaudal trend of development. The first two years are marked by a growth spurt, which is an accelerated increase in weight and height. The average weight of a neonate is about 3.3 Kgs (7-7.5 lb) and the average length is 20 in or 51 cm. At 4-6 months babies will double their weight and add about 4 inches. By twelve months, the child will triple their weight. Length is about 20-29 inches. By twenty four months, the weight is four times, and the length is about 32-36 ins.

6.1.2 Development of the Brain and the Nervous System

At birth, the brain is about 25% its eventual adult weight. At 1 year it is about 66%, 2.5 years 76% and 5 years 90%. The last three months of pregnancy and the first two years

after birth is a brain growth spurt. Proper nutrition -- that is adequate and balanced diet is crucial at this stage. Different parts of the brain develop at different times. The earliest to develop are the parts that control primary bodily movements and the primary sensory areas such as vision and hearing. The parts that direct the various thinking processes develop later.

Myelination (the development of sheaths around individual nerves, which insulate them from one another) follows a chronological sequence similar to the one of the brain. Myelination enables messages to pass down the nerves easily. At birth the spinal cord is not fully myelinated. The sheath covers first the nerve fibers in the head, shoulder region, arms, hands, upper chest, abdomen, and legs then feet. So infant will lift head and neck before trunk. By 2 years, myelination is almost complete.

6.1.3 Skeletal Development

At birth the bones are soft and easy to break. They are too small and flexible. The bones harden gradually following the Cephalocaudal trend of development. For example the bones of the skull mature first, followed by those of the hands and finally those of the legs. Foods rich in mineral salts are important.

6.1.4 Muscular Development

At birth, the muscle tissue is 35% water. Muscle fibers begin to grow as the cell fluid in the muscle tissue is replaced with protein and salts. Muscular development proceeds in a Cephalocaudal direction. Maturation occurs gradually over childhood but accelerates at adolescence.

NB:

A child's ability to perform a physical task depends not only on the maturation of the neurons in the brain, but also in the maturation of the muscle and skeletal systems. Such a state of maturation is known as *readiness*. Until children reach a state of readiness they will be unable to perform a task, even with training or practice.

6.2 PHYSICAL BEHAVIOR – MOTOR DEVELOPMENT

Motor development involves development of two tasks: -

1. Locomotion - involves body control and movement. The following are age norms (in months) for important motor milestones when 90% of the children have mastered the skill:

Table 6.1: Appearance of important motor milestones.

Skill	Month when 90% have mastered the skill
Lifts head while lying on stomach	3.2
sits propped up	4.2
Sits without support	7.8
Stands holding on	10
Stands alone	13
Walks well	14.3
Walks up steps	22
Kicks ball	24

2. Prehension - Involves reaching out activities, holding, seizing and grasping. These activities involve eye-hand co-ordination. They begin with the grasping reflex and uncoordinated arm –movements.

At 1 month, the infant will not grab an object. Will just stare at it.

At 2 months, will grab an object placed on the hand.

3 months they begin to reach out for objects

4 to 6 months will grasp objects but sometimes cannot let go

7 months they can pick objects with their fingers. They begin by using their palms and fingers then use the index finger with the side of the palm.

9 months, they can pick with their thumb and forefinger.

6.3 IMPORTANCE OF MOTOR DEVELOPMENT

1. Stimulates intellectual growth and creativity
2. Encourages socialization, independence, and self-confidence.
3. Promotes good mental health
4. Strengthens self concept

6.4 FACTORS AFFECTING PHYSICAL AND MOTOR DEVELOPMENT

1. **Heredity** -- determines the rate of growth and maturation, bone age, height etc.
2. **Nutrition** - should be balanced and sufficient otherwise will lead to stunted growth, kwashiorkor, rickets, marasmus obesity etc.
3. **Diseases** – prolonged illness interferes with steady growth
4. **Child rearing practices**- practice, stimulation
5. **Stress** - affects hormonal secretions causing physical dwarfism.

Implications

For proper physical and motor development, the following factors need to be considered:

- Importance of maturation

Importance of individual differences

- Importance of adequate and balanced diet
- Medical attention and immunization
- Hygienic conditions
- Provision of a rich environment
- Encouragement of play and manipulation of objects.

Summary



In this lesson, we have discussed changes in weight and height. brain, skeletal and muscular development; the progression of motor development and finally factors influencing physical motor development. Implications for optimal physical and motor development were also discussed.

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Review Activity



- 1) Identify children at different age groups in your community and describe their physical and motor changes and characteristics.
- 2) Explain any factors that may have contributed to their growth and development.

LESSON SEVEN: PERCEPTUAL DEVELOPMENT**7.0 INTRODUCTION**

Are newborns born with certain perceptual capacities or are they acquired through learning experience. What can newborns perceive?

This lesson discusses perceptual development in children. First we will look at the definitions and theoretical explanations; perceptual changes in perceptual development; depth perception; object and visual perception and finally, we will look at factors that influence perception.

OBJECTIVES

By the end of this lesson, the learners should be able to:

- a) Differentiate between sensation and perception.
- b) Explain the empiricists and natiristic approach to perceptual development
- c) Discuss changes in perceptual development
- d) Discuss depth, object and visual perception
- e) Explain the factors that affect perceptual development.

7.1 DEFINITION AND THEORIES OF PERCEPTUAL DEVELOPMENT

Sensation - Refers to the process through which information about the environment is picked up by sensory receptors and transmitted to the brain. Infants have sensory abilities, that is, they respond to light, sound, smell, touch and taste.

Perception: Refers to the interpretation of the nerve impulses we receive through the sense organs into recognizable patterns such as objects, words, smells etc.

7.2 PERCEPTUAL DEVELOPMENT

There are four main theories of perceptual development: -

1. Traditional theories

They believe that our perception of objects is developed through learning by associating the multiple sensations that an object evokes. For example in perceiving an orange, one has to perceive and associate all the related parts.

2. Gestalt Psychology

They claim that we perceive objects by organizing principles that result from the brain's natural organizational processes influenced by innate principles. They believe that the brain has an innate ability to organize objects as wholes and not parts – through the principles of proximity, similarity, continuity and closure.

3. Empiricists

They argue that infants are born “blank slates” then experiences are imprinted on them. Later, they learn to discriminate between sensory inputs. Thus, perception develops as a result of a long learning process.

4. Nativity's

Argue that many perceptual abilities are present at birth due to structural characteristics of the nervous system. Infants try to create order and organization in their perceptual world. According to Eleanor Gibson (1969) a child perceives an object when he or she detects its unchanging feature, noting the arrangement and movement of its surfaces. According to Eleanor Gibson, perceptual development is a process by which one learns to make increasingly fine and complex discriminations. At first, a baby perceives only gross differences For example between sound and silence. With experience, a child learns how to discriminate For example between musical sounds and other sounds, then will learn how to discriminate between one note and another.

Like adults children are exposed to a continual flow of perceptual stimulus, but they do

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not perceive every stimulus. Their perception of the same stimulus changes with age. According to Gibson and Spelke (1983), although perceptual development appears continuous, five changes can be seen:

1. As children grow, their perception becomes more selective and more purposeful. They focus on stimuli that have more functional value.
2. Children become increasingly aware of the meaning of their perceptions, For example whether pleasurable, useful or painful.
- 3 Perception becomes more sensitive as children develop. They begin to detect increasingly subtle aspects of stimuli For example the banging of doors meaning annoyance.
4. Children become more efficient in picking up critical information from stimuli, For example an ambulance siren, a gunshot.
5. Children become more proficient at generalizing perceived meanings from one situation to another.

At birth, both perception and sensation are present. Newborns see, hear, smell and taste but most of their sensory abilities are immature. Their perception is very selective, For example will pay attention to bright lights, loud sounds, and objects within a foot of their eyes. The sense of touch is more developed at birth. Vision is the least developed at birth. At three months, they begin to respond to mother's facial expressions. At four months, they begin to use both eyes better.

Hearing is quite acute. Sudden noises startle newborns and make them cry. Rhythmic sounds For example a lullaby or heartbeat soothes them and puts them to sleep. By one month, they can distinguish mother's voice. By four months, they can identify the voices of the most familiar people in their lives.

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By one year, the physical senses are almost fully developed. The eyes can focus accurately, they can properly register line and color, can distinguish variations in pitch and sound, and the nervous system can carry signals promptly -to the coned receiving center in the brain.

During the pre-school years, perceptual awareness develops to the point at which initial perceptions usually trigger a chain of complex mental processes aimed at problem solving or at acquiring a fuller understanding of the situation. The Child's perception becomes finer, sharper and more reliable.

Perceptual development is closely linked to motor development. The increased mobility exposes the child to a larger environment with more objects to perceive, more action to perform, and hence more opportunity for schemata to be formed. A favorable environment rich in intellectual as well as emotional stimulation is a important factor in the full development of the child's potential

7.3 DEVELOPMENT OF DIFFERENT TYPES OF PERCEPTION

1. Object Permanency

This is the realization that objects continue to exist even when out of sight. The process is as follows: -

0-4 months: A child acts as though object does not exist if covered up or disappears from view.

4-8 months: Begins to search for an object she or he saw disappear. If one drops a cloth on an object, the child withdraws the hand without the object.

8-12 months: Will search for an object that has disappeared. But if it is hidden again in a new place, will look in the original place.

12-18 months: Will search for an object that has disappeared. But if it is hidden when not looking, will not look further.

By 24 months: Child has object permanency. Will search for an object everywhere.

2. Development of Depth Perception

Depth perception is the ability to judge relative distances. Campos et al noted that every young child (up to 5 months) may show interest in depth cues, but do not fear until 6 months of age.

Gibson and Walk (1960) carried out an experiment on depth perception. They used the visual cliff, which creates an illusion of depth. It consists of a raised platform, half of which is patterned (checkerboard surface) and half is a glass. Under the glass part, a sharp drop in the patterned surface produces the illusion of a cliff. Infants are placed on the patterned side and called to the deep side.

Results

Very young children up to 5 months look intently at the illusion cliff. Heart rate decreases showing that they perceive the difference are puzzled but they do not cry. Older children, 6 – 8 months will not cross even after feeling the solid glass below and seeing their mother the other side. 9 months – their heart rate speeds up – afraid.

NB:

Depth perception is important because it warns babies of imminent danger for example falling off tables or chairs. Babies who have depth perception will sit on the edge and cry for help.

3. Development of Picture Perception

According to Piaget, children's recognition of pictures as real is not predetermined though it is innate. Dominant features in the picture first capture their attention.

According to Spelke, babies learn how to differentiate pictured objects at the same time they learn the distinctive features of real objects.

At six to seven years, children can act mentally upon the picture and not just the dominant features. They can analyze it, integrate its features and reverse figure and ground at will. Judging depth in pictures depends on learning because the child has to disregard information that betrays the flat nature of the picture. Older children are

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aware of movement in drawings such as leg not on ground but do not perceive it when indicated by clouds of dust or lines. By 12 years they are able to detect motion in both.

4. Development of Perceptual Constancy's

Perceptual Constancy's – treating things as unchanging.

- a) Shape Constancy – begins at 3 months and develops slowly. When shown irregular shapes, treated them as new. Improves with learning and maturation.
- b) Position Constancy - from 6 months. Improves with maturation, experience and learning.
- c) Size Constancy – occurs when someone walking away seems smaller but sees him same size though the retinal image becomes smaller. Size constancy begins at 5 months and develops steadily with time. By 6 – 8 months has size constancy.

7.4 FACTORS INFLUENCING PERCEPTUAL DEVELOPMENT

- i. Motivation / Need
- ii. Expectation
- iii. Genetic – maturation and experience
- iv. Culture
- v. Context
- vi. Stimulating – rich environment

7.5 Summary



- ✓ Perception is the interpretation by the brain of sensory input.
- ✓ Infants are born with a range of perception, which improves with learning experience.

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Review Question



Discuss the role of environment and heredity in perceptual development.

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LESSON EIGHT: COGNITIVE DEVELOPMENT

8.0 INTRODUCTION

Cognition is a mental activity through which human beings acquire and process information and knowledge so as to understand the world. In this lesson, we will examine Piaget's theory of cognitive development and its implications to teaching and learning. Finally we will look at factors that influence cognitive development.

OBJECTIVES



By the end of this lesson, the student should be able to:

1. Define the terms/concepts cognition and cognitive development.
2. List and explain the major characteristics of and achievements in each of Piaget's stages of development
3. Discuss various factors that influence cognitive development
4. Explain how Piaget's theory can be applied in educational settings.

8.1 DEFINITION OF TERMS

Cognition: The activity of knowing, or the mental processes by which knowledge is acquired.

Operations: Mental skills or actions (compare with physical actions).

Cognitive Development: Refers to the changes that occur in children's mental skills and abilities over time such as attending, perceiving, learning, thinking and remembering.

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Cognitive Equilibrium: Piaget's term for the state of affairs in which there is a balanced or harmonious relationship between one's thought processes and the environment.

Schema / Schemata / Cognitive Structure: A basic unit of knowledge, action, belief or image. Piaget's term for the patterns of actions or mental structures that are involved in the acquisition of knowledge. For example a child forms schemes of things that can be sucked, grasped etc.

Reversibility: The ability to reverse or negate an action by mentally performing the opposite action

Object Permanency: Understanding that objects continue to exist even when out of sight.

Adaptation: One's inborn tendency to adjust to the demands of the environment.

Assimilation: The process of interpreting new experiences by incorporating them into existing schemata. For example sees horse and refers to it as doggie.

Accommodation: The process of modifying existing schemata in order to incorporate or adapt to new experiences. Realizes differences of horse and doggie that is create new schemata.

Conservation: The recognition that the properties. For example weight, length, volume of an object or substance does not change when its appearance is altered in some superficial way.

Centration: The tendency to focus on only one aspect of a problem when two or more aspects of a problem at a time.

Decent ration: The ability to consider more than one aspect of a problem at a time.

Primary Circular Reaction: A pleasurable response centered on the infant's own body that is discovered by chance and performed over and over again.

Secondary Circular Reaction: A pleasurable response centered on an external object that is discovered by chance and performed over and over again.

Tertiary Circular Reaction: An exploratory schema in which the infant devises a new method of acting on objects to reproduce interesting results.

Hypothetical Deductive Reasoning: A style of problem solving in which the possible solutions to a problem are generated and then systematically evaluated to determine the correct answer.

8.2 PIAGET'S STAGES OF COGNITIVE DEVELOPMENT

Piaget identified four major periods (stages) of cognitive development. According to Piaget, these stages form invariant developmental sequences that all children progress through the stages in precisely the same order. There can be no skipping of stages because each successive stage builds on the accomplishments of previous stages. There are also individual differences in that environmental influences may accelerate or retard a child's rate of development. So the age norms are only rough approximations. The stages are as follows: -

The differences in stages are both qualitative and quantitative.

1. **The Sensory - Motor Stage (Birth to 2 years):** - The dominant structures at this stage are behavioral schemata or action schemes (compare concepts in older children and adults) which evolve as infants begin to co-ordinate their sensory input and motor responses in order to "*act on*" and get to know the environment. During this time, infants evolve from reflexive creatures into reflective playful problem solvers who have learned

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a lot about themselves and their environment. The following are some important attainments at this period:

a). Growth of problem solving skills: -

During the first month, children's activities are confined to exercising their innate reflexes such as sucking on objects.

The first non- reflexive schemata emerges at 1-4 months of age as infants discover by chance that they can emit and control various responses For example sucking thumbs, blowing bubbles etc. They realize these behaviors are satisfying and worth of repetition. These simple repetitive acts are called **primary circular** reactions and are always centered on the infant's body. They are called primary because they are the first motor habits that appear and circular because the pleasure they bring stimulates their repetition.

Between 4 — 8 months infants also discover that (also by chance) that they can make interesting things happen to external objects For example making a rubber duck quack by squeezing it. These responses are called secondary circular reactions and are also repeated for the pleasure they bring.

8 - 2 months truly planful responding first appears as infants begin to co-ordinate two or more actions to achieve simple objectives For example if you place a toy under a cushion the child may try to lift the cushion with one hand while using the other to grab the toy. This is an intentional schema in which two unrelated responses, lifting and grasping — are co-ordinated as a means to an end.

12 — 18 months, infants begin to experiment with objects (trial and error experimentation). They will try to invent totally new methods of solving problems or reproducing interesting results For example instead of just squeezing a rubber duck to make it quack, may decide to step on it, crush it to see whether these actions will have the same effects. These trial and error exploratory schemata are called tertiary circular reactions. They signal the emergence of true curiosity.

18-24 months children begin to internalize their behavioral schemata to construct mental symbols or images, that is, inner experimentation. They become able to solve problems without resorting to trial and error activities. For example, a child who cannot reach a toy on a table decides to use a stick to bring the toy near. The *“problem solving”* occurs at an internal symbolic level.

b) Development of imitation: - There is

- i). Voluntary Imitation: - becomes more precise at age 12-18 months.
- ii). Deferred Imitation: -the ability to produce the behavior of an absent model. First appears at 18 — 24 months. This is because now they are constructing mental symbols or images of a model’s behavior that are stored in memory and retrieved later to guide the child’s recreation of modeled sequence. For example, temper tantrums. But other investigators claim that deferred imitation appears much earlier 9-14 months (Meltzoff 1985,88).

- c). Object Permanence:** Out of sight is no longer out of mind. One of the more notable achievements of the sensory — motor period is the development of the object permanency — the idea that people, places and things continue to exist when they are no longer visible or detectable through other senses. By 18-24 months, object permanency is complete.

2. PRE -OPERATIONAL STAGE (2-7 YEARS)

During this stage children become more proficient at constructing and using mental symbols to think about the objects, situations and events they encounter. Piaget divides this period into two sub stages: the pre-conceptual period (2-4yrs) and the intuitive period (4-7 years).

a) The Pre-Conceptual Period (2-4yrs)

This period is marked by the appearance of the symbolic language function. The ability to make one thing — a word or an object stands for, or represent something else. For example, words come to represent objects, persons, events. The child can now

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reconstruct and talk about the past and talk about items that are not present. Pretend play also blossoms at this time. Will pretend to be mum, dad, baby, doctor etc. and will use any props to symbolize other objects For example shoe box to represent baby's crib, piece of wood to represent lorry, corn husk baby etc.

Contribution of play to Cognitive Development: -

- Permits children to practice their competencies in a relaxed and care free way
- Helps to nurture curiosity object permanence, inner experimentation etc.
- Helps children to develop additional cognitive and social skills
- Helps in the resolution of conflicts and in the compensation of unsatisfied needs and the inversion of roles For example obedience and authority.
- Enables children to try out roles that other people play while encouraging them to think about the feelings of the individuals who actually live these roles. (Role Taking)

Limitations in the pre-conceptual reasoning: -

- i. Animism: attributing life and life like qualities For example motives, feelings, intentions) to inanimate objects. For example don't throw the stone it will feel pain, the sun is angry that is why it is hiding.
- ii. Does not separate fantasy from reality. Dreams considered real.
- iii. Transductive reasoning: reasoning from the particular to the, particular. When two events occur closely together, the child assumes they are casually related, that is, one caused the other. For example napping in the afternoon, if misses the nap in the afternoon then it isn't afternoon.
- iv. Egocentrism: a tendency to view the world from ones perspective and to have difficulty recognizing another person's point of view. They are not aware that others may have ideas that are different from theirs For example when telling a story, will leave important information as if one was there or will ask, "didn't you see?" Speech is not adapted to listener.

b) The Intuitive Period (4-7 Years)

The child's thinking is called "intuitive" because his understanding of objects and events is "centered" on their single most salient perceptual feature. For example, color. They focus on the way things appear to be — rather than on logical or rational thought processes. The limitations of a perceptually based, intuitive logic is apparent when 4-7 year olds work on class inclusion problems that require them to think about whole /part relations.

- They do not understand the difference between a class and a representative of that class.
- Their thinking is centered on one salient feature and fails to consider other features.
- They cannot conserve. This is because they are incapable of mentally reversing or undoing the flow of action, argument and therefore do not realize that the liquid in the short, broad container would attain its former height when poured back into a tall thin container. They are also unable to attend simultaneously to both height and width. They are not able to decentre.

3. The Concrete Operational Stage (7-11 Years)

This stage is called concrete operations because Piaget believes that children at this stage can apply their operations only to objects, situations and events that are real or imaginable. Some achievements of this stage are: -

a). Class inclusion

They understand part or whole relationships and the difference between a class and a representative of the class.

b). Conservation

They can conserve liquids, mass, number, volume and area.

c) Mental representation of actions.

The concrete operational child can construct accurate mental representation of a complex series of actions. For example, can sketch a map of the route to school while a pre-operational child may not.

d). Relational Logic.

One of the hallmarks of operational thinking is a better understanding of relations and relational logic. Concrete operators are capable of seriation, an operation that enables them to arrange a set of stimuli along a quantifiable dimension such as length. A related ability is the concept of transitivity, which describes the relationships in serial order. For example, if John is taller than Sam who is taller than Mark, then John must be taller than Mark.

The transitive inferences (thinking/logic) of concrete operators are generally limited to real objects that are physically present. Cannot apply this logic to verbal problems or to abstract signifiers such as the X's & Y's in algebra.

4. THE FORMAL OPERATIONAL STAGE (12 AND BEYOND)

This is the last of Piaget's intellectual stages. Thinking is no longer tied to the observable or imaginable. Formal operators can now reason quite logically about abstract ideas that may have no basis in reality. They enjoy thinking about hypothetical and are likely to develop some very unusual and creative responses. Their approach to problem solving becomes increasingly systematic and abstract - like the hypothetical - deductive reasoning of a scientist.

In summary, formal operators can solve complex verbal problems involving the future, the past and the present. Their thinking is rational, systematic and abstract. Can think about thinking and operate ideas as well as tangible objects and events.

Limitations

At the beginning of this period, ego-centrism in the form of naïve idealism is very high and affects their reasoning capabilities. Naïve idealism is the inability of the adolescent to differentiate between his own idealistic thoughts and the "real world". The criterion for making judgments is what is logical and not what is realistic.

Another characteristic of this ego-centrism is personal fable — belief that one is unique and has a personal mission in life so problems cannot befall them. For example,

pregnancy, death, aids etc.

Imaginary audience is the third characteristic. This is a belief that those around are focusing on oneself like an imaginary audience watching caused by extreme consciousness. Feels that everyone is watching what one says and does. These three forms of egocentrism decrease as one encounter “hard reality”.

8.3 FACTORS INFLUENCING COGNITIVE DEVELOPMENT

1. Biological factors — maturation of nervous system
2. Socio-cultural factors social interaction, educational factors/schooling, and cultural factors
3. Experiences in the physical environment- rich and poor environments, mental stimulation

Application of Piaget’s theory to teaching and learning

1. When to teach, what to teach. Curriculum materials to consider child’s level of development. Readiness is important.
2. How to teach- teaching methods and materials consider child’s level of development. Child should be actively involved- interacting with objects and materials that is learning resources. Children should be encouraged to discover.
3. Encourage peer interaction — reduces egocentrism.
4. Consider individual differences because children because children develop at different rates.

8.4 Summary



- ✓ Children are active processors of information.
- ✓ Cognitive development follows an invariant sequence of stages.

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Review Questions



- 1) Discuss the various stages of Piaget's theory to teaching and learning.
- 2) Discuss factors from your local community that may affect children's cognitive development.

LESSON NINE: LANGUAGE DEVELOPMENT**9.0 INTRODUCTION**

Language is used through out the human life span for communication; thinking; problem solving; creative activities and writing. Language acquisition significantly alters the child's cognitive and social experiences and helps children to form a view of themselves. In this lesson we will trace the path of language development. This will be followed by theoretical explanations of language acquisition, and finally factors influencing language development.

OBJECTIVES

By the end of this lesson, the student should be able to:

- 1) Define different terms in language development
- 2) Describe the stages of language development.
- 3) Discuss and evaluate the basic features of theories of language acquisition
- 4) Discuss factors influencing language development

9.1 DEFINITION OF TERMS

Language - Language is a system of communication within a particular society or group. Human language uses sounds to transmit meaningful communication.

Language has several functions:

- Communication of thoughts, desires, and needs to others.
- Translation of experiences into symbols so that one is able to remember the experiences better.
- Helps to increase understanding.
- Social interaction.

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- Transmission of culture

Acquisition - The process by which we identify, perceive, register and record information in our memory.

Holophrastic stage - A period in which a child speaks in one word “sentences” usually between 12-18 months.

Telegraphic speech - Very abbreviated sentences consisting of two words, which correctly follow the structure of language.

Motherese - A simplified way of speaking, which facilitates language learning

Over-regularization - A process of making irregular word forms

Pragmatics - The study of how language is used in a social context.

Over-extensions - Process of using words to cover more than the adult meaning

Morpheme - The smallest unit of a language that by itself has a recognizable meaning. It is a word or part of a word.

Syntax - Consists of rules by which words and morphemes are combined to form longer units such as clauses and sentences

Semantics - Is the study of meaning of how the sounds of language are related to the real world and our own experiences.

Phonology - Is the study of the sound system of a language and how it develops.

9.2 PROCESS OF LANGUAGE ACQUISITION

The process of language development follows the following stages:

1. Pre-speech Communication (birth to 12 months)

- Before the development of speech, babies cry to communicate hunger, discomfort, or pain. Parents interpret their cries and respond to them. Other forms of pre-speech communication include the following. Cooing appears from about three to five weeks. These are repeated vowels-like sounds such as ooh!, ah! Vocalizations. From four to six months, babbling begins. This involves a mixture of vowels and consonants e.g. ma/ma, pa- pa.
- From ten months, babies use both gestures (point) and changes in pitch to communicate certain emotions, questions, requests or commands.

It is important to note that children understand language before they are able to speak.

2. Holophrastic Stage (10 to 18 months)

Children begin to utter single words to refer to specific tangible things they can see. A single word could be used to mean a lot of things; for instance:

“Door” could be used to mean:

That is a door

Is that a door? (Asking a question)

Open the door (state a demand).

The one word can only be understood in context. The listener has to infer the child’s intentions from the intonation and the specific situation. The language is said to over-extend e.g. a child will call all animals are doggie.

Overextension is as a result of limited vocabulary. The process provides an opportunity for the child to learn additional vocabulary.

3. Telegraphic speech (18 to 30 months)

Children begin to put two words together e.g. baby chair, more chair, throw ball – like a telegram. These sentences contain only the critical content words,

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leaving out the articles. The rudiments of grammar start to appear. Two-year olds continue to use intonation to indicate meaning.

4. Acquiring complex rules (2 years to pre-adolescent)

Children learn syntax (grammatical rules of language) by building on their knowledge of what words mean. Initially, the rules are over applied, for example when they learn the past tenses of verbs, they seem to apply them in all cases e.g.

make - maked

hold - holded

keep - keeped i.e. they over apply the rules for making past tenses. This is called over regularization. By three years, children have a vocabulary of 900 words. As they begin to make sentences, they discover the rule of combining words (syntactical rule). Errors are common in their speech and this provides an opportunity for correction.

9.3 THEORIES OF LANGUAGE ACQUISITION

There are four theories that explain the process of language acquisition. These are the learning theories, biological theories, cognitive theories and the interactionist theories.

1. Learning Theories.

According to this theory, children learn language because adults reinforce correct usage of their language. Successive approximations are rewarded or reinforced. Children also learn to imitate the language and accent of members of their social group. Children who are talked to more learn more vocabulary and those who are corrected do better than those who are not.

2. Biological (Innate) theories

The sequence of language acquisition is broadly similar in all societies. It occurs in all human cultures and all languages have certain features in common. Human beings have an innate device, the Language Acquisition Device (LAD) without which language cannot develop.

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1. Cognitive Theory

According to this theory, children have the ability to categorize their world even before they can communicate with others in language. They understand and use particular linguistic structures only when their cognitive abilities enable them to do so. Children form schemes to explain events in their lives and only then talk about them. As they develop more intellectually, they produce elaborate sentences that are expanded by adults and older children.

2. Social Interaction Theory

This theory deals with social skills of children and the effect of social environment. It emphasizes the child's early experiences of communicating and interacting socially with people in their surroundings. Adults give meaning to sounds and utterances of infants through comments, interpretations, extending and repeating what the child says.

9.4 FACTORS INFLUENCING LANGUAGE DEVELOPMENT

1. Social environment. The environment needs to be stimulating in terms of language acquisition. There should be suitable role models and regular use of language.
2. Presence of adult speakers rather than other children.
3. Cognitive development
4. Maturation and other developmental factors such as hearing, moth deformities etc.
5. Bilinguals.

9.5 Summary



There are our main stages in language acquisition

✓ Theoretical approaches to language were discussed.

Language acquisition is influenced by an interaction between innate and environmental factors.

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Revision Questions



- 1) What is the role of parents/caregivers in language development.
- 2) Explain the various theories of language acquisition
- 3) Describe the stages of language development.

LESSON TEN: SOCIAL DEVELOPMENT**10.0 INTRODUCTION**

We are social beings that depend on others for our needs. Social development is a process through which we acquire the ability to behave in accordance with social expectations, play approved social roles, develop appropriate social attitudes and acquire a sense of self-identity. In this lesson we will look at various theories of social development; the process of social development; the role of play in social development; and finally factors influencing social development.

OBJECTIVES

By the end of this lesson, the student will be able to:

- 1) Describe the main theories of social development.
- 2) Explain the process of social development.
- 3) Discuss the roles played by various agents of socialization.
- 4) Discuss play and its role in social development.
- 5) Explain the factors influencing social development.

10.1. THEORIES OF SOCIAL DEVELOPMENT

There are three theories that explain social development.

1. The psychoanalytic theory.

According to this theory, social development is influenced by the development of sex typed behaviors, which are acquired when children learn to identify with the same sex parent after resolving the Oedipus and Electra complexes.

2. The social learning theory.

Children develop socially through observation and imitation of others in the immediate environment. Those observed may be parents, siblings, teachers and peers. Appropriate

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imitation is reinforced. For example, boys are reinforced for aggression while girls are reinforced for submission and dependence. Inappropriate behaviors are punished thus reducing the chances of being repeated.

3. Cognitive theory.

This theory emphasizes the role played by mental processes in understanding one's sex and gender. Gender identity begins by realizing that one is a boy or girl by noting the anatomical differences between boys and girls. The child then labels themselves as male or female and begins to act accordingly. If this does not happen, one may experience gender identity problems such as transsexuals.

10.2 THE PROCESS OF SOCIAL DEVELOPMENT

From birth to six months, the neonate is asocial that is insensitive to other people. It is self-centered and this is because it has little communication.

From six to twenty months, the child begins to appreciate what other people do for them such as feeding, care and warmth.

From two to five years, the child is gradually separated from the mother and begins to interact with other children and siblings.

From six to eleven years, the child spends more time with peers in school. There are both formal and informal interactions. The associations are basically with the same sex. From twelve years onwards is the onset of adolescence. The peer group becomes more important and they begin to associate with members of the opposite sex.

10.3 AGENTS OF SOCIALIZATION

The following are the main agents of socialization.

The Family

Parents train their children in the following areas: eating, toilet training, sex modesty, dressing and tidiness, respect for adults and those in authority and cultural inhibitions.

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Children are trained through rewards and punishments and by imitation of role models.

The school

Children learn how to relate to others formally and informally. The school imparts knowledge and skills and proper attitudes to pupils to help them fit in the society. The school also reinforces the values of the society. For example emphasizing cleanliness, co-operation, respect etc.

Peer groups

Children learn meaning of role, status and leadership from peers. The peer group also influences fashions, habits and trends. From the peer group, children acquire the following benefits:

- It provides opportunity to practice leadership skills, which may enhance a child's self-esteem necessary for a healthy personality development.
- It teaches the child a variety of social skills such as cooperation and sharing.
- It provides an emotional outlet and social emotional support when one is experiencing problems.
- It provides opportunity for one to express him among equals without fear of judgment or reprimands.

Social organizations

These include religious organizations, clubs, societies etc.

Mass media

Includes electronic and print media. Their role is to educate, inform and influence. The mass media can enhance prosocial behavior, language and cognitive development. It can also introduce anti social behavior such as aggression and gender and racial stereotypes.

10.4 THE ROLE OF PLAY IN SOCIAL DEVELOPMENT

In playing, children learn to combine freedom with personally imposed rules in activities that have no goal outside of the activity itself. Play involves freewheeling fantasy, while games have externally imposed rules, they are usually competitive and have winners and losers. Young children engage in play, older children engage in games. Play socializes and educates youngsters.

Styles of play**1. Non-social play (2-3 years)**

It includes

- Unoccupied play –time is spent observing some event or object.
- Onlooker play – involves watching other children play and not joining in except to ask a question or suggestion
- Solitary independent play – involves playing with oneself without interacting with others.
- Parallel play – playing by oneself but by other children who are also playing.

2. Social play (4-5 years)

Includes

- Associative play – playing with others in a common activity, sharing play materials and influence each other's play activity, however there is no coordination or division of labor or unified goal.
- Co-operative play – more organized and has a common goal and common rules.

NB: social play is declining due to;

- Influence of the television
- Solitary and educational toys
- Growing use of computers by the children.
- Families with fewer siblings.
- Lack of space etc

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Types of play

There are four types of plays;

1. Functional play

Most common in the first and third years and most common in boys than girls. It involves simple repetitive muscular activities (with or without objects) For example pulling at toys.

2. Constructive play

Begins at the age of 2 years. Most common between the age of 4 and 6 years. It involves manipulation of object to build or create something. For example using blocks, clay etc. Girls engage in more constructive play than boys. Constructive play aids in problem solving by increasing flexibility and consolidating learning, encouraging elaboration and enhancing creativity.

3. Dramatic play(pretend or fantasy) involves either using imaginary situations or characters. It may be simple or elaborate. For example playing careers using puppets or costumes. Dramatic play allows learning through role-play, permits children to rehearse problems in a no failure situation, and lets children feel more powerful and competent by taking on the roles of parents, workers and even super heroes. Pre-school children can use fantasy play to handle feelings of helplessness, frustration and anger.

4. Games. For older children. Includes activities involving externally imposed rules, structures and a goal. For example playing hopscotch, marbles etc. Replace dramatic play and is common in ages 6-12 years. Gives practice in following complex rules, cooperating in large and small groups, dealing with winning or losing.

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Importance of Play

1. The physical activity of play helps young children to develop and improve their mental skills.
2. Imaginary and fantasy play is a step towards the development of abstract thinking.
3. Play helps children to persevere and build attention skills.
4. Play is important in social-emotional development.
 - Children use play to understand, express, share and control emotional experiences.
 - Allow children to get rid of aggression and to learn to control aggressive urges.
 - Through play children form their first friendships and build social skills.
 - Provide a fairly safe environment in which children can compare their own behaviors and skills with those of other children.
5. Helps in muscular coordination and motor skill are refined.
6. Increases self-confidence.
7. Develops cognitive, social and emotional skills for example problem solving, language competence and interaction.
8. Shows child's inner needs and desires.
9. Helps children to better understands themselves, others and events.
10. Learns how to formulate, negotiate, argue and follow rules.
11. Improves creativity imagination, and tolerance.
12. Helps children to acquire new information and understanding.
13. Learns how to listen to other's views.
14. Learns to communicate ideas and acquire more vocabularies.
15. Resolve conflict- endurance, sympathy, and self-control, give and receive.
16. Low impulsivity, low aggression, sharing, cooperation, independence, social maturity.

10.5 FACTORS INFLUENCING SOCIAL DEVELOPMENT

1. Presence of a handicap
2. Heredity
3. Culture

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4. Appropriate role models and socializing agents.
5. Disciplinary practices at home and school
6. Birth order
6. Rate of maturation

10.6 Summary



- ✓ Socialization equips children with a sense of identity
- ✓ Social development occurs in stages.
- ✓ Various theories of social development exist.
- ✓ Many agents influence socialization.
- ✓ Play is an important aspect in social development.

Review Questions



1. Discuss the three theories of social development.
2. Describe the stages of social development
3. Discuss the role played by the various agents of socialization.
4. Explain how play contributes to other aspects of development.

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LESSON ELEVEN: EMOTIONAL DEVELOPMENT

11.0 INTRODUCTION

Emotions are what make us seem most human. We rage; laugh, cry, fear and we love. Having feelings is an important part of being human. Do babies have feelings? There is reliable evidence that young infants are emotional creatures. In this lesson, we will discuss development of emotions; expression of emotions; recognition and interpretation of emotions; emotional attachment; and finally factors influencing expression of emotions.

OBJECTIVES



By the end of the lesson, the learner should be able to:

1. Explain what emotions are.
2. Discuss the development of emotions.
3. Explain the factors that influence the expression of emotions.
4. Describe how children recognize and interpret emotions in other people.
5. Distinguish between the different types of attachments and their effects on later development.
6. Discuss factors promoting and inhibiting attachment.

11.1 DEVELOPMENT OF EMOTIONS

Various emotions appear at different times over the first two years of life.

At birth, babies show interest, distress, disgust and contentment

From two and half to seven months, emotions such as anger, sadness, joy, surprise and fear emerge. These emotions up to seven months are called primary (or basic emotions). They are present at birth or emerge early in the first year. They are believed to be biologically programmed, for they emerge in all normal infants at roughly the same ages

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and are displayed and interpreted similarly in all cultures (Camras *et al* 1992; Izard 1982; 1993).

From two years, infants begin to display emotions such as shame, embarrassment guilty; envy; and pride. These emotions are called secondary (or complex) emotions. They are sometimes called self-conscious emotions, because each involves some damage to or embarrassment of our sense of self. They depend in part on cognitive development (self-recognition) and the ability to evaluate ones performance as good or bad.

However, children are likely to display self-evaluative emotions when someone else is present to observe their behavior. It is usually at elementary school period that children feel socially prideful or shameful about their conduct in the absence of external surveillance.

11.2 EXPRESSION OF EMOTIONS

Each society has a set of emotional display rules that specify the circumstances under which various emotions should or should not be expressed. For example, American babies learn that intense emotion is okay as long as it is positive, whereas Gusii babies learn to restrain both positive and negative emotions (Dixon *et al.* 1981; Levin and Levin 1988).

To comply with a culture's emotional display rules, the child needs to acquire the following skills:

- a) Emotional self-regulation. This includes strategies for regulating and controlling emotions, or adjusting emotional arousal to a comfortable level of intensity.
- b) Acquiring emotional display rules. This involves suppressing the unacceptable emotion one is experiencing and replacing them (outwardly) with whatever feeling that the display rule calls for in that situation. For example, acting happy after receiving a disappointing gift. This is involves emotional deceit.

As early as two years, children have the same emotions as older children and adults.

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However, what differs is their expression of these emotions. At infancy, inner feelings and outward expressions are clearly matched. In general, two year olds express their emotions in an immediate, impulsive and direct manner. They cannot wait to have their needs satisfied. They cry, throw temper tantrums, hit out until they have their needs satisfied. Pre-school children begin to mask their feelings, but in most cases, they wear their feelings on their face and express them freely.

During the elementary school years, children become more aware of socially sanctioned display rules, that is, what emotions to express and which ones to hide in particular situations. They are more verbal and thoughtful in their expression of emotions. They take longer to build and express their emotions. In addition, their expression is more diversified and less direct. For example, they learn to displace their anger For example kick a chair or a younger child.

Factors influencing expression of Emotions

1. Interaction with parents, other relatives and peers
2. Informal teaching. For example when boys cry, they are scolded and told that men don't cry but girls are tolerated even encouraged to cry.
3. Reward and punishment
4. Modeling
5. Direct instruction
6. Culture

11.3 RECOGNIZING AND INTERPRETING EMOTIONS

At three months, babies can discriminate their mother's happy, sad or angry expressions when the facial configurations are accompanied by a happy, sad or angry tone of voice, but they become gleeful in response to a happy expression and distressed by their mothers anger or sadness.

Eight to ten months, infants' ability to interpret emotional expressions becomes more obvious. They begin to monitor their parent's emotional reactions to uncertain situations

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and then use this information to regulate their own behavior. This ability is called **social referencing** and increases with age and is soon extended to strangers as well (Feinman,1992).

By twelve months, infants will approach and play with unfamiliar toys if a nearby stranger is smiling, but will avoid these objects if the stranger displays a fearful expression.

By two years, children will look to their companions after appraising a new object or situation. This shows they are using others' emotional reactions to assess the accuracy of their own judgments.

Two to three years, children become better at interpreting others' emotions. This is enhanced by the ability to talk about emotions in family conversations.

Four to five years, children can explain why playmates are happy, sad or angry, but they focus more on external events as causes than on internal needs, desires moods or motives.

Six to nine years, children achieve important breakthroughs in emotional understanding. For instance:

- They rely more on internal and external information to interpret emotions
- They recognize that many situations elicit different emotional reactions from different individuals
- A person can experience more than one emotion
- They display some ability to integrate contrasting facial behavioral and situational cues to infer what the emotions might be. These abilities are helped by decent ration.

11.4 EMOTIONAL ATTACHMENT

Attachment refers to the formation of an enduring social-emotional relationship between an infant and another person. Bowlby (1969) used the term to describe the strong affect ional ties that bind a person to an intimate companion. The first attachment is always between the infant and the mother/ caregiver. There are 3 qualities of attachment:

1. Secure attachment — Infant explores while alone with the mother and is visibly upset by separation. When the mother returns, infant is happy and welcomes physical contact with her. The child is outgoing with strangers when mother is present.
2. Insecure attachment (anxious and resistant) — Child appears anxious and are unlikely to explore while mother is present. They become very distressed when the mother departs. When mother returns they are ambivalent, that is they are unlikely to initiate contact.
3. Insecure attachment (anxious and avoidant) — Uninterested in exploring when alone with their mothers. They show little distress when separated from the mother and will avoid contact with her when she returns. They are not particularly wary of strangers.

Stages of Attachment (Schaffer and Emerson, 1964)

1. The asocial stage (0—6 weeks)

The baby is “asocial” in that many kinds of social and unsocial stimuli produce favorable reaction, and few produce any protest.

2. The stage of indiscriminate attachments (6 weeks - 7 months).

They enjoy human contact but they are somewhat indiscriminate.

Will protest if any adult puts them down or leaves them alone.

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They enjoy attention from anyone.

3. The stage of specific attachment (7 months)

They begin to protest if separated from one particular individual, usually the mother. They begin to fear strangers. This shows they have formed their first genuine attachments.

4. The stage of multiple attachments

They become attached to other people - father, siblings, grandparents or a regular baby sitter.

Factors promoting attachment

- a) Parents begin to prepare long before the baby is born, For example planning and excited about the development of the baby.
- b) Feeding and overall responsiveness and amount of stimulation.
- c) Warmth and contact are important.
- d) The appearance of the baby -large forehead, chubby cheeks and soft rounded features. They appear cute and lovable.
- e) Level of intellectual development, particularly the ability to discriminate familiar persons from strangers.
- f) In born programmed responses that enable them to promote interactions from which attachments are likely to develop. e.g. rooting; grasping; smiling and crying.

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Factors That May Inhibit Attachment

- a) Unlovable babies — irritable and unresponsive. For example premature babies are less alert and avoid caregiver's attention; children born to drug addicts are more irritable, withdrawn, sluggish and unresponsive.
- b) Unreachable caretakers — if depressed, pregnancy was unplanned, or following preconceived notions of child—rearing; if they are emotionally insecure, For example feelings of being unloved, neglected or abused as children.
- c) Environmental factors — too many children to look after and no help, unhappy marriage — depression. Unhappily married parents are often stressed to the point that it is difficult for them to respond warmly and sensitively to their infants.

Effects of attachment on later development

- a) If securely attached the child will be more curious; interested in learning; cooperative; friendlier toward adult more liked by peers; better at problem solving; and more complex and creative in symbolic play during the infancy stage.
- b) A securely attached infant derives comfort from close companions and can use them as safe bases for exploration.
- c) In nursery school, those who were securely attached become social leaders, they often initiate activities; they are more sensitive to the needs and feelings of other children; they are popular with peers and they are described as curious, self—directed; eager to learn; and less dependent on adults.
- d) Insecurely attached infants do not venture far from their attachment object

though they derive very little comfort or security from them. They are usually hostile and aggressive and are likely to be rejected by peers.

- e) In nursery school, the insecurely attached are socially and emotionally withdrawn; are hesitant to engage other children in play activities; and are described by observers as less curious, less interested in learning and much less forceful in pursuing their goals

11.5 IMPORTANCE OF EMOTIONAL DEVELOPMENT

Ability to understand and interpret others' emotions and to express emotions is an important aspect of social cognition. It enhances social development in the following ways:

1. Emotional expressions at infancy help infants and their close companions "to get to know each other". For example cries of distress — attention, smile — social, fear and sadness — insecure, anger — stop what you are doing, joy - prolong whatever interaction is going on.
So infant emotions promote social contact and help caregivers to adjust their behavior to the infant's needs and goals.
2. Ability to interpret others' emotions enables one to enjoy good relations with peers thus becomes better in social competence.
3. Social referencing enables children to acquire knowledge. For example, a mother's pained expression and accompanying vocal concern may suggest that the knife in ones hand is dangerous.
4. The information contained in caregivers' emotional displays contributes to the child's understanding of the world in which he lives.

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11.6 Summary



- ✓ Different emotions appear at different times
- ✓ Each culture has a set of emotional display rules, which specify under which circumstances various emotions should or should not be expressed.
- ✓ The ability to recognize and interpret other's emotions enhance social development in various ways
- ✓ The type of attachment an infant forms with another person determines later development.
- ✓ Attachment is enhanced by both innate and environment factors.

Review Questions



1. Discuss the various types of attachment and their effects.
2. Explain factors that promote attachments in babies.
3. What is the importance of recognizing and interpreting emotions correctly in others

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LESSON TWELVE: MORAL DEVELOPMENT

12.0 INTRODUCTION

Moral values are social values. They deal with individual behavior in relation to fellow human beings. Moral development can be defined as a process through which an individual acquires the ability to distinguish between right and wrong. In this lesson we will discuss Piaget & Kohlberg's theory of moral development. Finally we will look at factors influencing moral development.

OBJECTIVES



By the end of this lesson, students will be able to:

1. Discuss Piaget and Kohlberg's theories of moral development.
2. Discuss various factors that influence moral development

12.1 PIAGET'S THEORY OF MORAL DEVELOPMENT

Piaget was the first to attempt a systematic Study of children's moral reasoning.

Working with children between 3 and 13 years, Piaget investigated moral thinking of children in three areas: -

- The formulation of rules
- The relation of intention to the attribution of culpability
- Justice

Piaget observed children at play; participated in the game of marbles with them; interviewed them about the rules of games, the nature of Justices; and the morality of characters in stories. The two of the stories are presented below.

Story A

A little boy who is called John is in his room. He is called to dinner. He goes to the

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dinning room, but behind the door there was a chair and on the chair there was a tray with 15 cups on it. John didn't know what was behind the door. The door knocked the tray and all the fifteen cups got broken.

Story B

Once was a little boy whose name was Henry. One day when his mother was out, he tried to get some jam out of the cupboard. He climbed on a chair and stretched out his arm. While trying to get it, he knocked over a cup. The cup fell down and broke.

Who was more naughtier and why? Well Intentioned John who caused much damage or ill intentioned Henry who does damage? On the basis of children's responses, Piaget identified three broad stages of moral development: -

1. Pre-moral (birth – 5 years)

Piaget noted that during the preschool years, children show little understanding of social rules. They invent their own rules; change rules at will and will conduct the game according to their own private desires and fantasies.

2. Stage of heteronomous morality or moral realism (5-10 years)

From around five years, children believe rules from authorities are fixed, sacred and absolute. They cannot be changed. They believe Justice is subordinate to adult authority. For example it is never right to tell a lie because the parents say so.

They also believe in immanent Justice, a kind of inherent retribution that is breaking of rules will surely result in punishment. They Judge actions by looking at the consequences rather than the intentions, For example in the above stories, they will say John is naughtier because he broke more cups.

3. Stage of Autonomous Morality or Moral relativism. (10 - 13 years)

They see morality as more relative to the situation. In judging whether a particular action is right or wrong, they consider intention as well as consequences. They realize

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that agreement or consensus can change rules. They realize that disobedience; lying and violation of rules are sometimes justified. They also realize that not all wrongdoing will be punished. Piaget believed that, change from moral realism to autonomous morality is due to greater cognitive abilities and socialization with peers.

12.2 KOHLBERGS THEORY OF MORAL DEVELOPMENT

Kohlberg expanded piaget's work. Working with the pre-adolescents, and adults Kohlberg presented them with moral dilemmas embodying a conflict between several contending claims for justice. He presented them with moral dilemmas to elicit their reasoning. One of the famous dilemmas was the Heinz dilemma produced below.

In Europe a woman was near death from a special kind of cancer. There was one drug that the doctors thought that it might save her. It was a form of radium that a druggist from the same town had recently discovered. The drug was expensive to make but the druggist was charging ten times what the drug cost him to make. He paid \$2000.00 for the radium and charged \$2000.00 for a small dose of the drug. The sick woman's husband, Heinz went to everyone that he knew to borrow money, but he could only get together \$1000.00 which is half what it cost. He told the druggist that his wife was dying and asked to sell it cheaper or let him pay later. But the druggist said "No I discovered the drug and am going to make money from it. So Heinz got desperate and broke into the man's store to steal the drug for his wife. Should the husband have done that? Why?

From their responses Kohlberg identified three levels and six stages of moral development.

I) Pre-conventional Level

Emphasis is on avoiding punishment and getting rewards. There are two stages in this level.

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Stage 1: Punishment-obedience orientation

‘Good’ behavior is defined by a desire “to avoid punishment” imposed by some external authority. For example will say that Heinz should not steal the drug because he might be sent to jail if caught.

Stage 2: Instrumental-relativistic orientation

The right thing is that which satisfies ones own needs. The right is that which will make one get rewards. There are elements of fairness and reciprocity but for physical pragmatic reasons. Moral action involves making deals for fair exchange (Market place morality). For example he should not steal the drug because his wife may die when he is in jail, and it won’t do him any good.

II) Conventional level

Emphasize on social rules. Stages three and two are in this level.

Stage 3: Interpersonal concordance of good boy/girl orientation

Good behavior is that which pleases others and is approved by them. There is much conformity to stereotypical images of what is majority or “natural behavior”. A person lives up to what is expected of people in your role as son, brother, etc. for example, Heinz should steal the drug because if the wife dies people may think that he is an inhuman husband.

Stage 4: Authority and social order maintaining orientation.

Right behavior consists of doing one’s duty, showing respect for authority and maintaining the given social order for its own sake. Believe in absolute worth of fixed rules. For example, Heinz should steal the drug because he made a marriage vow that he will take care of her, or he should not steal the drug because he will be breaking the law that it was made for everyone’s good.

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III) Post- Conventional level

Emphasis is on moral principles. Stage five and six are in this level.

Stage 5: Social contract legalistic orientation.

Right action is defined in terms of general individual rights and standards, which have been critically examined and agreed upon by the whole society. Recognizes that laws are formed by mutual agreement and can be changed through the democratic process. Also realizes that individual rights can sometimes supercede these laws if the laws become destructive. For example, Heinz should steal the drug. The fact that her life is in danger transcends every other standard you might use to judge his action. Life is more important than property.

Stage 6: The universal ethical principle orientation

This is the highest in Kohlberg's stages of moral reasoning. Individuals make their judgment on the basis of universal ethical principles. These principles are logically consistent and are based on the highest ethical values of justice, reciprocity, equality and respect for human life and rights. Moral action is prescribed by inner conscience and may or may not be in accord with public opinion or societal laws. So a person does the right things to avoid self-condemnation: "I couldn't leave with myself if I did that.

12.3 FACTORS INFLUENCING MORAL DEVELOPMENT

Cognitive development

For instance, with the decline of cent ration in middle childhood, children are able to consider simultaneously both consequences and intentions when judging morality of an act.

1. Increased social experience with peers.

This reduces unilateral respect for adult authority and increases respect for peers and their points of view. They also learn to take roles and assume another's perspective. Can be influenced to do good or bad.

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2. **The family** – through direct instruction, imitation, story telling, proverbs, rewards and punishment.
3. **Religious organization** – teaches moral / religious values
4. **Mass media** – may learn bad or good behavior through observation and imitation.
5. **Culture/community** – has their own laws, customs and social norms
6. **Encouraging Independence of adult constraint** – by discussing moral issues in a rational way and by emphasizing consequences of their actions and others.

Summary



- ✓ Piaget presents three stages of moral development while Kohlberg envisages three levels and six stages of moral development
- ✓ Various factors influence moral development

Review question



1. Describe the characteristics of moral reasoning at each of Kohlberg's three levels.
2. Explain how the school environment can influence moral development in children.

LESSON THIRTEEN: ADOLESCENT GROWTH AND DEVELOPMENT**13.0 INTRODUCTION**

The word adolescence comes from the Latin verb “adolescere” which means to grow into maturity. Hall defined it as a period in human development beginning at puberty and ending when full adult status has been attained by twenty two to twenty five years. In the African traditional setting, the adolescent period is marked by coming of age ceremonies. In this lesson we will address the following aspects of adolescent development: developmental tasks; developmental changes, which includes physical, cognitive, social and emotional aspects; challenges and problems facing the adolescents and coping strategies.

OBJECTIVES

By the end of this lesson, the learner should be able to:

1. State the developmental tasks of adolescence
2. Describe the physical changes at the adolescent stage and discuss their impact on adolescent behavior.
3. Describe the cognitive, moral social and emotional changes at the adolescent stage.
4. Discuss the challenges and problems facing the adolescents and what can be done to enhance their growth and development.

13.1 DEVELOPMENTAL TASKS

Developmental tasks are inescapable expectations or obligations that a person has to attain by the end of a specific stage of development. Durojaiye(1979) suggested eight developmental tasks that have to be satisfied during the adolescent period:

1. Attaining individuality
2. Making progress towards an organized personality pattern
3. Developing a philosophy of life

4. Developing a concept of values and desirable behavior
5. Achieving a place in society
6. Understanding personal assets and liabilities
7. Maturing of plans for future living
8. Establishing deep personal relationships with individuals of both sexes.

Progress towards the attainment of these tasks ensures that boys and girls move from childhood to adulthood mentally, emotionally, socially and physically. They also attain the attitudes and beliefs needed for effective participation in society. In the following sections we are going to see some of the changes that take place during this period.

13.2 PHYSICAL GROWTH AND DEVELOPMENT

The onset of adolescence is heralded by two significant changes in physical development:

1. The adolescent growth spurt, that is a rapid acceleration in weight and height that marks the beginning of adolescence.
2. Puberty - derived from the Latin word “pubertas” which means age of manhood or literally to grow hairy. It refers to that point in life when we reach sexual maturity and become capable of producing a child. The period marked by the onset of the maturation of the reproductive functions is referred to as **pubescence**. It lasts for two years ending in puberty.

It has been established that at some biologically determined time, the pituitary gland sends a message to a young person’s gonads or sex glands (ovaries and testis). The timing is regulated by the interaction of genes, the person’s health, nutrition and other environmental factors. The sex glands then produce hormones, which stimulate sexual maturation (estrogen, testosterone and androgen).

The average age for the onset of pubescence for girls is ten years with puberty following at age twelve with a normal range of nine to sixteen years. For boys the average age is

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twelve to fourteen years with a normal range from eleven to eighteen years. Marked physical changes take place during this period:

Primary Sex Development

These are changes related to the maturation of the external and internal sex organs. For example, in males, the penis, scrotum, testes, seminal vesicles and the prostate gland mature. In females, the ovaries, fallopian tubes, uterus, vagina, and breasts also reach maturity. The most dramatic sign of sexual maturity in girls is the “**menarche**” i.e. the onset of menstruation. It occurs about two years after breast and uterine growth begin to grow and after a girl’s height has slowed down. It occurs between eleven to sixteen years. The presence of sperms in the male’s urine is considered the principle sign of sexual maturation.

Secondary Sex Characteristics

These are the features that accentuate the anatomical distinction between boys and girls. For example, in males the voice becomes hoarse, wet dreams, hair in armpits and genital areas, facial hair, broadening of shoulders etc. In females, growth of breasts, widening of the pelvis (hips) and pubic hair.

Growth Spurt

There is an accelerated rate of increase in weight and height. Girls reach their adult height by fourteen or fifteen years and most boys by eighteen. There are also changes in the body proportion. The legs and arms lengthen hips in girls become wider and boys develop round shoulders.

Physiological Changes

There are greater changes in the internal systems of the body. Respiratory, circulatory, digestive, blood pressure and pulse rates reach their full growth.

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Physical Activity and Ability

There is a greater increase in muscular strength. Males have larger muscles, lungs, and greater capacity for carrying oxygen in the blood. This accounts for their considerable greater strength and endurance. Boys are better in activities that involve speed and muscular strength, while girls are better in jumping and throwing.

Cognitive Development

The middle part of the brain spurts out and reaches full development at this period

Psychological and Social Impact of the Physical Changes

Adolescence can be a very embarrassing time. This is because young people are convinced that everyone is watching their every move and their bodies are constantly betraying them. These changes have significant behavioral implications.

1. If they are not informed of the changes in advance, it can lead to extreme anxiety, withdrawal and depression.
2. They become very sensitive about their appearance e.g. fat, acne, big breasts etc.
3. They worry a lot. For example, girls worry about menstrual accidents while boys may worry about uncontrolled erections. They may become very apprehensive about participating in activities such as standing in front of the class to read.
4. They become very self-conscious and feel that everybody is watching their every move.

Early and Late Maturation

Young people vary widely in the age at which they reach puberty. For example, at fifteen years, a girl may still be flat chested and very small while another at the same age may be fully developed and able to reproduce. Therefore, some adolescents mature early while others mature late due to individual differences. These variations are normal and do not either help or interfere with the eventual achievement of full physical and sexual

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maturity. However, they can affect the way adolescents' view themselves and the way they are viewed by others. The following are some possible effects of either early or late maturation.

Effects of early and late maturation.

BOYS

Research has shown that early maturing boys are more poised, relaxed, good natured, popular with peers, likely to be leaders and less impulsive than late maturers. They also have a high self-esteem.

Adults and peers rate Early maturers as physically more attractive, more composed, and more socially sophisticated than late maturers. They are independent, self-controlled and dominant.

Early maturing boys may have problems in living up to others' expectations. They are expected to act mature than they are capable of.

Research has found that late maturers feel more inadequate, rejected, dependent, more aggressive, and insecure. They are more likely to rebel against parents and think less of themselves. They may feel and act more childishly.

Late maturers have the advantage of having a longer childhood whereby, they don't have to deal with the new and difficult demands of adolescence.

GIRLS

Early maturing girls have been found to be less sociable, expressive, poised and more introverted and shy.

Early maturing girls make better adjustments in adulthood.

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Late maturing girls are initially more gregarious, socially poised, assertive, active, more popular with peers, and more satisfied with their body image. However, at late adolescence, the early maturing girl becomes more popular with both sexes, more self-poised, better at coping, more self-directed cognitively, socially and emotionally.

13.3 COGNITIVE AND PSYCHO-SOCIAL DEVELOPMENT

In this section, we are going to discuss the changes and characteristics of adolescents in the following aspects of development: cognitive, moral, social and emotional.

13.3.1 Cognitive Development at the Adolescent stage

Most of the adolescents are at the Formal Operational stage although some may still be at the concrete operational stage. Characteristics of thinking at this stage are:

- They can reason about abstract ideas. They are freed from concrete reasoning. They can think about the future, and can think in terms of symbols and imaginary events.
- They acquire the ability to generalize facts - deductive reasoning, see relationships and to solve problems of increasing complexity and difficulty.
- They have the ability to reason hypothetically, e.g. what if; supposing...They can generate very unusual and creative responses. Concrete operators are more likely to balk at hypothetical propositions and say it can never happen.
- They have the ability to appreciate metaphoric meaning, i.e. using a word or phrase with one literal meaning to describe another object or event. For example, “a blanket of snow”. “Shattered faith”.
- Their approach to problem solving becomes increasingly systematic and abstract, like the hypothetical deductive reasoning of a scientist.
- They have a tremendous development of long-term memory and imagination. They also develop an increase in language. They have an increased ability to communicate with other persons.

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- They can identify with conditions and characters in the larger world.
- Argumentativeness. They have the desire to practice their newfound mental ability - to consider various points of view. It helps to stretch out their reasoning ability. They need to be engaged in discussions.

Adolescent Egocentrism

In spite of the tremendous development in their cognitive ability, adolescent thought is limited by adolescent egocentrism. There are three forms of this egocentrism.

Personal Fable.

Adolescents see themselves as much more central and significant on the social stage than they actually are. For example, they believe that no one else has ever had the particular emotional experiences they themselves are having, e.g. so angry, so much in love etc. An example is a teen telling the mother “ you don’t know how it feels to be in love”. They also see themselves destined for great fame and fortune e.g. discovering a cure for cancer or being a great philosopher.

Imaginary Audience.

They agonize over the fact that others are constantly watching and making comments about them. They fantasize how others will react to their appearance and behavior. For instance, they will spend hours before a mirror thinking others will judge the final result. They will enter a room regarding themselves as the most attractive and admired human being alive. If they have a slight blemish, they will not want to go to school, or they will wish they were invisible. They keep seeing disapproval everywhere.

Invincibility Fable.

The feeling that they are somehow immune to the laws of mortality and probability. They believe they have a mission in life and problems cannot befall them yet. They feel

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magically protected from harm such as death, accidents, pregnancies etc. This leads to risk taking behaviors e.g. drunken driving, promiscuity etc.

Naive Idealism.

Inability to differentiate the ideal from the practical. They imagine an ideal world and they feel compelled to try and bring reality closer to the ideal. For example, they have perfect answers to social problems. They point out all the shortcomings in people and things and finds fault with almost everybody especially those in authority.

13.3.2 Adolescent Moral Development

According to Kohlberg, most adolescents are at the conventional level (stage three and four), but a small percentage especially the delinquents are at the pre-conventional level. A few may be at the post-conventional level. Rewards and punishments guide pre-conventional reasoning on right behavior; conventional - conforms to the laws of society and post - conventional they follow their own principles.

Most adolescents are at stage three, which is characterized by mutual and interpersonal expectations, relationships and conformity. What is right is living up to what is expected by people close to you, which generally involves conformity to stereotypical good behavior. Anticipation of approval or disapproval by significant others and the need to be a good boy/girl guides behavior.

At stage four, what is right is fulfilling the actual duties to which you have agreed. Right is maintaining the society, group or institution. One does the right to avoid the breakdown of the system or the institution “if everyone did it”.

Due to advances in cognitive development, the adolescent critically examines the moral code and asks a number of questions. This makes the adolescent to achieve autonomous moral reasoning. They internalize moral concepts and are able to choose what is right or wrong by themselves.

13.3.3. Social Development at the Adolescent stage

Physical changes at adolescence have an effect on social development and relationships.

The appearance of adult physical characteristics leads to demands from adults that the adolescent should act more grown up. This may lead to friction between the adolescent and the adults around him. Adolescence is a period of ambivalence towards adults especially teachers and parents whom the adolescent may think are old fashioned and unreasonable. Let us examine a few aspects of social relationships at this stage.

1. Parent - Adolescent Relationships

The family is both a source of security and a difficult problem for the adolescent. It is a source of security because it offers a refuge, a place of acceptance and a place where material needs are provided e.g. food, clothing, shelter, clothes. It is a problem because it is a barrier to free choice of activities, friends, subjects etc. This leads to difficult relationships between parents and adolescents. Parents and teachers need to enhance smooth relationships with adolescents by doing the following:

- Parents and teachers need to appreciate the difficulties of this period. They need to provide abundant love and understanding.
- Involve the adolescents in decision-making. There is a need to tone down parental domination. Identity cannot be achieved if parents are too domineering.
- Avoid dictating to them. Explain your reasons.
- The school should provide opportunities to foster social relationships and development, e.g. social functions such as games, debates, seminars, excursions and trips.
- Adolescents' needs to trust parents in order to accept their decisions as wise and just.

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2. Peer - Adolescent Relationships

The adolescent develops more relationships with people outside the family, especially the peers (people of the same age and the same behavioral level). The peer group is important because it gives opportunities to the adolescent to practice social skills, conversation, courtesy, and cooperation and leadership roles. Major functions of the peer group are:

- It provides a source of information and comparison about the world outside the family.
- The adolescent receives feedback about his abilities, e.g. whether good in sports, popular with peers etc.
- Creation and reinforcement of values and attitudes
- Provides moral and emotional support and alternatives
- Can try possibilities safely without blames, and rebukes by adults. Can rehearse roles and test out ideas and behaviors, e.g. information about sex, co-operative and competitive behavior among equals, sex-role behaviors and the expression of aggression and play.
- Provides refuge from adult pressures and demands
- Provides real acceptance and approval

NB:

Peer pressure and peer conformity may discourage individuality and self-assertion. Negative peer pressure is more effective on adolescents from unhappy homes.

13.3.4 Emotional Development during the Adolescent stage.

The adolescent experiences similar emotions as the young child, but there are differences in the amount, intensity, types of responses and types of stimuli that creates the emotions of the adolescent. The physical changes at puberty make their emotions change so frequently that they have been described as ambivalent, or vacillations. Adolescent emotions have also been described as heightened. This can be explained by the following factors:

Causes of Heightened Emotions**Parent - adolescent relations**

Excessive parental domination; unreasonable restrictions; sibling rivalry etc. Feels parents are unreasonable and old fashioned. Leads to friction. They feel like children in decision-making, and when they become emotionally volatile, parents reprimand them for being childish. This makes the adolescent confused.

Adjustment at home and at school

The changeover to new patterns of habits creates emotional tensions. The process of changing from total dependence to independence roles with greater responsibility disturbs the adolescent.

Unfavorable relations in the home

Lack of proper guidance in preparation for the problems of adolescence; lack of understanding their interests and points of view.

Social Expectations

They are expected to think and act like an adult for whom he is not physically and intellectually ready.

Adjustment to members of the opposite sex

There is attraction towards the members of the opposite sex but the adolescent is not able to understand the correct social behavior

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Religious conflicts

He critically examines his parent's beliefs and begins to criticize them

School failure

Failure in school causes heightened emotionality so much so that a number of adolescents commit suicide leave home or withdraws from school.

Vocational problems

They develop anxiety when they see many unemployed. They worry about their future occupation.

Characteristics of Adolescent Emotions

Complexity

Adolescent emotions become complex due to the experiences he gets in his environment. He learns to conceal his true emotional experience.

Development of abstract emotion - They can express their emotional feelings in relation to objects which are abstract or which are not present in concrete form.

Emotional feelings are widened - Can relate to future, past and present and can be relate to many people.

Bearing of tensions - They develop competencies to bear tensions in different social situations, i.e. develops emotional self-control. They feel an inner freedom to feel and experience in an intimate personal way.

Capacity of shearing emotions - They develop the capacity for the concern of the feelings of others, and an increased capacity for sharing emotional experiences with others. Begins to love his neighbor as himself.

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Loyalties expand - The adolescent emotional world expands beyond home and the neighborhood and he identifies with peers and leaders of various fields -heroes.

Realism in emotional experience - Recognizes the weakness and the strength of ones character. He enters the period of reality.

Reviewing of hopes and aspirations - It is a time when one has high hopes and aspirations for his future life. It is a time when one works realistically to achieve these aspirations.

Toleration of loneliness - Sometimes they like to be left alone

The adolescent is able to project his feelings on others.

Increased compassion - This is the ability to enter into kinship with the feelings and the impulses involved in any sort of emotional experience whether it is joy or sorrow.

Common Adolescent Emotions

Anger-moodiness

Is the most disruptive while moodiness is least disruptive. Caused by restraints and disruptions. Manifested verbally and through name-calling.

Jealousy

Occurs when the adolescent feels threatened or insecure because his relationship with another is threatened in some way. Is expressed through verbal sarcasm, ridicule, derogatory comments and making another person look bad.

Envy

May be envious of persons who have material possessions and social status. May be the cause of juvenile delinquency.

Fear, worry and anxiety

Fear is less recognized than before. The adolescent may plan to avoid embarrassing situations. Worry and anxiety are imaginary fears which may include feelings of inadequacy, problems related to school performance, physical appearance, social acceptability, sex, marriage, home situation etc.

Affection

Common in early adolescence although it is usually flirtations. Later, affection develops by concentrating on one member or a small group.

Happiness

Feels happy when at ease, achieves a sense of superiority over others and when releases stored up emotional energy.

13.4 CHALLENGES AND PROBLEMS FACING ADOLESCENTS.

The adolescent period can be a very trying and traumatic period for some of the adolescents. Common adolescent problems include juvenile delinquency, suicidal tendencies, substance abuse, chronic aimlessness eating disorders and truancy. In addition, the adolescent is faced by a number of challenges, which include resolving of identity issues, sexuality related challenges, school and career-related challenges, choice of friends, peer pressure and conflict with adults. The following are suggestions that can help the adolescent to cope with the challenges and problems of this period:

1. Provide a favorable home environment where there is unconditional acceptance and love, and respect. Parents also need to model responsible behavior and learn to listen to what the adolescents have to say. In addition, provide basic necessities.
2. Develop accountability and responsibility in the adolescents. This can be achieved through discussion of conflicting values, giving them experience in decision-making, giving them opportunities to do things for themselves and letting them experience the consequences of their behavior.

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3. Provide continuous guidance and counseling as the need arises. Prepare them for the changes before they occur. Give them information so as to make informed choices and decisions.
4. Encourage them to discover their talents and support them. Identify skills they can train in especially during school holidays.
5. Encourage delay of gratification and self-control. Let them participate in community service and other sublimation activities.
6. Parents and teachers to tone down their domineering attitudes. Involve them in decision-making and discussion of various issues affecting them. As much as possible use the democratic process or the participatory approach.
7. Train them in life skills such as decision-making, assertiveness and choice making. Encourage high self-esteem and positive self-concept.
8. Provide emotional support and intervention for those who may face problems. For example, provide suicide awareness programs, psychotherapy, drug therapy and peer counseling to those who may be in a crisis.

13.5 Summary



- ✓ Adolescents experience physical changes, which have major effects on their personality and behavior.
- ✓ Adolescents are faced with various challenges and problems
- ✓ Various strategies and skills can be used to help them to cope.

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Review Questions



Identify the challenges and problems facing adolescents in your local community.
Discuss what can be done to help them go through adolescence smoothly.

LESSON FOURTEEN: ADULTHOOD DEVELOPMENT**14.0 INTRODUCTION**

In the earlier lessons we focused on changes in growth & development in childhood and adolescence. These changes can be viewed as a preparation for the period of adulthood or maturity. The period of adulthood is the beginning of a commitment to careers, marriage and parenthood. However, there are variations based on factors such as cultural demands, social-economic classes, nationalities and ethnicity, personality differences and historical events such as wars. In this lesson, we will discuss adulthood in three stages: early adulthood, middle adulthood and late adulthood and aging. For each of the stages we will look at the developmental changes, developmental tasks and problems or challenges at each stage.

OBJECTIVES

By the end of this lesson, the student should be able to:

1. Discuss developmental tasks and changes at each stage.
3. Discuss some of the challenges in adulthood
3. Suggest coping strategies.

14.1 EARLY ADULTHOOD (20 – 40 YEARS)**Developmental Changes****a) Physical Characteristics**

Strength, energy and endurance characterize Early adulthood. The peak of muscular strength occurs at about 25-30 years. The senses are most effective at this time.

b) Cognitive Characteristics

The main feature of adult thinking appears to be its practical nature, which allows for intellectual contradictions to exist side by side in a person's mind in the furtherance of a practical goal. Adults focus their intellectual energies on solving real problems. They accept contradiction, imperfection and compromise as part of adult life.

Developmental Tasks of early adulthood

Early adults are at Eriksson's stage six of intimacy versus isolation. Developmental tasks include the following:

1. Selecting a mate.
2. Marriage and learning to live with a marriage partner.
3. Starting a family.
4. Rearing children.
5. Managing a home.
6. Getting started in an occupation.
7. Taking a civic responsibility.
8. Finding a congenial social group.
9. Separating from parents.
10. Being more independent financially and psychologically.

14.2 MIDDLE ADULTHOOD (41-65 YEARS)

Developmental Changes

a) Physical characteristics

A number of physical changes take place during this period.

1. There is a marked and gradual decline in many aspects of physical development. Vision and hearing begin to decline.
2. Incidence of cardiovascular disease increases during this time. Blood pressure usually rises and the coronary arteries that supply blood to the heart begin to narrow during this time. These changes are linked to the lifestyle of a person.

3. Muscular strength declines steadily during this period.

4. **Menopause**

It is defined as the end of menstruation, which occurs as a result of ovarian degeneration and a decline of estrogen secretion. It occurs over a prolonged period of time from around 45 – 65 years. Major changes include: -

- i. Aging of ovaries and decline in output of ovaries i.e. the ova, estrogen and progesterone. This leads to inadequate uterine stimulation, which consequently leads to the end of menstruation and discomfort when having sex.
- ii. Hot flashes, headaches, dizziness, heart palpitations and joint pains.

NB: There are individual differences in the effects of menopause. Some feel it is the end of their sex life while others feel its liberating because there is no fear of pregnancy.

Male Climacteric syndrome

- i. Although men do not experience menopause, they experience age related changes in the reproductive system. Climacteric syndrome refers to the loss of the ability to reproduce. These changes appear from the 60's to the 70's. Characteristics include: -
- ii. Gradual decline in sperm production although viable sperms are produced by the oldest men.
- iii. Androgen (testosterone) levels gradually decrease and erection occurs more slowly leading to decreased sexual performance.
- iv. The seminal fluid becomes thinner and its volume decreases.

Midlife Crisis: This is a period of transition at about 40 to 50 years during which individuals take stock and a look at their lives and ask was it worth the trouble? They may feel effective and competent and at the peak of their powers. To others, it may be a

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painful process. It's usually caused by a shift in control. They accept that they have no control over factors such as disease, death etc.

Factors that may facilitate the crisis are divorce, children leaving home, menopause, infidelity, competition, impotence, job changes, death of a spouse, biological aging and severe illness. How people react to becoming middle aged or the above factors, may determine whether they are in a period of transition, reassessment or midlife crisis. If they are regarded as a normal part of life it will less likely lead to crisis.

Developmental tasks

Middle adults are at Eriksson's stage seven of generativity versus stagnation.

Developmental tasks include the following:

1. Achieving adult and civic responsibility.
2. Establishing and maintaining an economic standard of living
3. Assisting teenage children to become responsible adults
4. Developing adult leisure time activities
5. Relating to ones spouse as a person because children have grown, left and parents are left alone once again.
6. Accepting the physiological changes of middle age.
7. Revise and re-appraise the early adult era and modifying unsatisfying aspects of previous lifestyle.

This can be a stressful stage as others depend on them – younger and older generation – and they have nobody to depend on.

14.3 LATE ADULTHOOD AND AGING (65 AND ABOVE)

Developmental changes

a) Physical characteristics

There are several major changes at this stage.

1. The entire circulatory machinery is less efficient

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2. Sensory and perceptual functions decline. They have visual problems, hearing loss, sense of taste declines, and vestibular senses loose efficiency and touch and movement decline with age.
3. The skin looses its elasticity i.e. its folded and wrinkled. Hair becomes thinner and turns gray.
4. They shrink in size and stoop. Chemical composition of the bones changes causing an increased chance in bone fractures.

b) Cognitive changes

1. Lack of mental alertness. Loss of memory and forgetfulness, which is made worse by cardiovascular disease and senility.
2. Decline in creativity.
3. More effective though slow in their work due to accumulated experience and knowledge.
4. Has a large amount of wisdom. Should be encouraged to be active both physically and mentally.

Developmental Tasks

They are at Eriksson's last stage of integrity versus despair. Developmental tasks include the following:

1. Adjusting to decreased physical strength and health.
2. Adjusting to retirement and reduced income.
3. Adjusting to death of a spouse.
4. Establishing an explicit affiliation with ones age group.
5. Establishing satisfactory physical living conditions.

Social life is restricted by loss of work-mates, relatives, spouse, friends and poor health. So they mainly rely more on their families for companionship – may feel neglected and lonely at old people's homes.

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Death and dying: Epilogue

Typical reactions to impending death include: -

1. Denial
2. Anger – directed at death, against those not dying, the healthy and robust, dependants, what is left undone etc. they should not be taken seriously.
3. Depression – allow them opportunity to express their sorrow
4. Bargains for additional time
5. Acceptance
6. Hope

NB:

Age is not the major factor determining entry into these stages. For example, if a girl marries at 16 years and gets children will already enter into the early adulthood stage. If one is about to die at 45 years due to illness, they will enter into integrity vs. despair stage.

1.4 Summary



- ✓ Adulthood is a period of maturity, which is associated with the beginning, and ending of a career, marriage and parenthood.
- ✓ The period marks a peak in physical and cognitive development.
- ✓ Middle age marks a gradual decline in physical, social, personal and economic performance.
- ✓ Intellectual and cognitive functioning remains stable throughout middle age.
- ✓ In late adulthood there is a marked decline in the efficient functioning of the senses, cognitive abilities and physical activities.

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Review Question



Carry out a survey of the old people in your community. Identify their problems, challenges and coping strategies.

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